

Toward a Principled Bayesian Workflow in Semantics: A Tutorial

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Abstract

When researchers study semantic phenomena using Bayesian cognitive models, they must validate these models to ensure they accurately formalize the underlying theory and predictions. This is particularly important, as Bayesian models have gained increasing popularity in semantics in recent years, yet the question of how to validate these models has not been addressed in the literature. Here, we present an R and Stan-based tutorial on five model validation methods based on the principled Bayesian workflow in cognitive science (Gelman et al., 2020; Betancourt, 2018; Schad et al., 2021). We apply these methods to a cognitive model on semantic representations of natural function words. Specifically, we demonstrate how to implement and interpret (1) prior predictive checks, (2) computational faithfulness, (3) model sensitivity, (4) parameter recovery, and (5) posterior predictive checks. The tutorial includes path diagrams for an easier understanding of the methods, troubleshooting checklists to address undesired outcomes, and R and Stan code to make it easier to apply these methods to new situations.

1 Introduction

The field of semantics has shifted from focusing solely on logical theories (e.g., Barwise and Cooper, 1981; Peters and Westerståhl, 2006) to evaluating model predictions based on experimental data. This transition prompted researchers to reconsider how they conceptualize semantics in their work, particularly in light of new insights they gained from empirical studies. While logical theories treat semantics as deterministic, the response behaviour observed in experiments revealed a more complex picture. Specifically, individuals seemed to respond in a probabilistic rather than deterministic manner. Moreover, their responses are influenced by suboptimal information processing, which introduces additional variability and noise into the observed patterns.

To navigate this complexity, researchers have adopted a probabilistic perspective on semantics. Rather than seeking to identify general laws, research in experimental semantics aims to capture and investigate probabilistic trends across populations. This shift has led to the development of computational models that describe probabilistic processes through probability distributions, model response patterns using statistical functions, and account for noise with additional parameters. Among these approaches, Bayesian cognitive models have proven to be indispensable tools in experimental semantics and are now applied to a wide range of semantic topics (e.g., van Tiel et al., 2021; van Tiel, 2014; Wallsten et al., 1986; Newstead, 1988; Schmidt et al., 2009). But what exactly makes Bayesian cognitive models special?

In our opinion, the primary advantage of Bayesian cognitive models is their ability to describe behavior probabilistically by formulating prior distributions on model parameters. The necessity of formulating prior distributions within the Bayesian framework is often framed as a weak point (Gelman, 2008). However, as a field so rich in theories, researchers should embrace prior distributions as they allow them to directly incorporate into their model their preexisting knowledge, semantic intuitions, and theoretical expectations (Vanpaemel, 2010; Lee and Vanpaemel, 2018). Furthermore, Bayesian

cognitive models are highly flexible, a feature that is particularly relevant when analyzing complex data (e.g., [Rouder and Lu, 2005](#)). Beyond the use of prior distributions, these models can be easily extended by adding parameters, incorporating hierarchies, or structuring the model according to specific aims of the study (e.g., adding a pragmatic component to a semantic one; [van Tiel et al., 2021](#)). Finally, posterior distributions offer a natural measure of uncertainty, as the parameter estimates vary in width depending on the informativeness of the data.

Despite the popularity of Bayesian cognitive models in experimental semantics, researchers in this field rarely validate their models before applying them to empirical problems. However, if not properly validated, Bayesian cognitive models may exhibit unexpected and undesired behaviors. They may predict, for instance, behavioral patterns that deviate from theoretical expectations, or the sampling algorithm may produce prior distributions that do not align with those specified by the researcher. This may lead to incorrect inferences, especially when the goal of the empirical work is not only to estimate parameters but also to test predictions and hypotheses using Bayes factors ([Etz and Wagenmakers, 2017](#); [Kass and Raftery, 1995](#); [Jeffreys, 1935, 1939](#); [Ly et al., 2016](#)).

In this tutorial, we aim to highlight these challenges and provide guidance on how to address them effectively. The outline of the tutorial is as follows. First, we will present three applications of Bayesian cognitive models in the field of semantics and pragmatics. One of these applications, the use of Bayesian cognitive models to capture the meaning of quantifier expressions, then serves as a running example throughout the rest of the tutorial. The tutorial itself explains each validation step in detail and showcases how to interpret the results based on the running example. We implement all methods in R-Studio ([RStudio Team, 2020](#)) and Stan ([Stan Development Team, 2023](#)). All code snippets are attached in one RMarkdown file in the appendix and we keep referring to the relevant passages during the explanation of the process. To simplify this, the codes are in numbered chunks of the markdown file, which can be accessed in our Open Science Framework folder at: <https://osf.io/wzeca/>.

1.1 Applications of Bayesian Cognitive Models in Semantics and Pragmatics

To demonstrate the versatility of Bayesian cognitive models, we will briefly discuss some examples of their application. While an extensive overview of their applications is beyond the scope of this paper, we focus on three prominent examples: their use for researching generalized quantifiers, gradable adjectives, and the rational speech act model.

Generalized quantifiers. Generalized quantifiers express a quantity or an extent, including numerical quantities, proportions, comparative degrees, or others (e.g., *Few*, *More than Half*; see [Szymanik, 2016](#), for an overview). Traditional approaches capture a quantifier’s semantic representation using formal logic derived from Generalized Quantifier Theory ([Barwise and Cooper, 1981](#); [Peters and Westerstahl, 2006](#)). On this account, it is commonly considered, for instance, that the quantifiers *Most* and *More than Half* are truth-conditionally equivalent, as both refer to proportions greater than 50%. However, several empirical studies have found that these quantifiers may have distinct semantic representations ([Ariel, 2003](#); [Kotek et al., 2015](#); [Solt, 2016](#); [Ramotowska et al., 2023](#); [Denić and Szymanik, 2022](#); [Talmina et al., 2017](#)).

To study the differences between the two quantifiers (among other things), [Ramotowska et al. \(2024\)](#) proposed a Bayesian cognitive model to describe individuals’ response behavior. Their Bayesian cognitive model captures semantic differences between quantifiers across three dimensions: the meaning threshold of a quantifier, that is, the proportion at which the quantifier is considered true, the vagueness of the meaning boundaries, and response noise. These three model parameters allowed [Ramotowska et al. \(2024\)](#) to measure semantic parameters independently of response noise and therefore to detect real differences between quantifiers more easily. Furthermore, the hierarchical component added to the model was used to study whether and to what extent individuals differ in how they assign meaning to quantifiers.

[Ramotowska et al. \(2024\)](#) indeed found indications that *Most* and *More than half* carry distinct semantic meanings. Their results suggest that, at the population level, *Most* is associated with a higher meaning threshold than *More than half* (53% versus 50%), greater uncertainty around the meaning threshold and slightly higher response noise rates. Although their results suggest that the two quantifiers differ in their meaning thresholds at the population level, this difference does not

hold for every individual. Instead, there appear to be different clusters of subpopulations, with some perceiving the two quantifiers as truth-conditionally equivalent. This example is a demonstration of how Bayesian cognitive models can provide a nuanced analysis of behavioral patterns at both the population and individual levels and can help identify which behavioral patterns warrant further explanation in semantic theories.

Gradable adjectives. Subjective gradable adjectives are adjectives whose semantic representation depends on the noun they modify and whose meaning can vary in intensity (e.g., *Tall*, *Cold*, or *Expensive*). Here, too, logical theories have struggled to fully account for observed behavioral patterns. The formal semantics of gradable adjectives typically focuses on overarching mechanisms of how adjectives are interpreted relative to their comparison sets (e.g., through prototypes, reference classes, or definitions, [Schmidt et al., 2009](#)). Within these mechanisms, however, these approaches remain vague on what kind of internal computations individuals use to verify these adjectives ([Schmidt et al., 2009](#)).

To explain how individuals relate adjectives to their comparison sets, [Schmidt et al. \(2009\)](#) formalized threshold-based models and a category-based model as Bayesian hierarchical cognitive models to describe the cognitive processes when evaluating the truth conditions of the gradable adjective *Tall*. The proposed threshold-based model assumed that individuals have an internalized meaning boundary for each adjective against which they compare the object in question. If the object exceeds this boundary, the adjective is considered accurate. The category-based model (based on the rational model of categorization, [Anderson, 1991](#)) assumes that the semantic representation of an adjective involves comparing the object in question to standards that represent extreme points of the category. For instance, when an individual evaluates the expression “a tall building,” the object is compared to both the tallest building they can think of and the shortest building they can think of. The expression is deemed accurate if considered a more typical example of a strong standard.

Crucially, for each family of models, [Schmidt et al. \(2009\)](#) implemented multiple types of statistical functions individuals may use to arrive at a tallness judgment. For threshold models, these functions included computing a specific percentage or the number of standard deviations above the mean at which a noun would be considered tall. For category-based models, different clustering algorithms were implemented to separate short from tall items. [Schmidt et al. \(2009\)](#) were able to identify, among the candidate models, two models (one threshold-based model and one category-based model) that best describe the data. This study contributes to the field of semantics by providing insight into the internal calculations individuals use to make decisions. This example illustrates how Bayesian cognitive models can contribute to theory development. Since these models require explicit specification of each mechanism involved, such as the statistical functions that determine decision criteria—they help make semantic theories more precise.

Rational Speech Act. While quantifiers and gradable adjectives are typically included in the study of semantics, we would also like to introduce an example from the closely related field of probabilistic pragmatics. The rational speech act describes the interaction between the speaker and the listener as a recursive, probabilistic reasoning process. The key idea is that both the speaker and the listener are rational agents who use the basic form of Bayesian inference to optimize communication. That is, speakers choose words from a set of alternatives that maximize informativeness and listeners infer the speaker’s most likely intentions based on the word, context and prior expectations ([Frank and Goodman, 2012](#)).¹ In this discipline, too, an increasing body of literature provides evidence for systematic variability that can be accounted for in cognitive models; for instance, listeners vary in how they judge the probabilities of the state of the world given a speaker’s choice of an alternative ([Degen, 2023](#)).

In [Franke and Degen \(2016\)](#) study, individuals played a reference game either in the role of a speaker or a listener. In the game, listeners were required to infer which object a speaker refers to based on minimal descriptive information. To describe observed empirical data in a reference game, the authors formulated two competing Bayesian cognitive models to study whether there exist individual differences in the production and comprehension of language. The authors compared a simple model, which assumes a homogeneous population where all speakers and listeners follow the same set of rules, with a hierarchical model which assumes that speakers and listeners represent a mix of different reasoning types.

¹Please note that the Rational Speech Act framework has also been applied to subjects from semantics including quantifiers ([Franke, 2014](#)) and gradable adjectives ([Qing and Franke, 2014](#)).

Their results show that a hierarchical Bayesian cognitive model, which accounted for variability in participants’ reasoning strategies, outperformed the simple model assuming a homogeneous population, particularly in language comprehension. This example illustrates that population averages may not always adequately capture individual behavior. Instead, the added complexity of the hierarchical cognitive model allowed for a more accurate description of individuals’ reasoning types. Once again, Bayesian cognitive models proved to be a valuable tool for studying the complexities observed in empirical data and for accounting for these complexities by expanding the model structure.

1.2 Bayesian Model Validation

With the increasing adoption of probabilistic models in the field of semantics, we can expect that novel Bayesian cognitive models will be proposed in the future and existing models will be refined to meet field-specific task demands. However, certain risks are associated with the development of Bayesian cognitive models. For instance, the model may fail to generate predictions that align with domain expertise, or may be unable to re-describe the observed data; imposed restrictions on the parameters (e.g., range restrictions) may lead to observed prior distributions that deviate from the expected ones; or the algorithm used to produce posterior samples may not function as intended. To guide researchers in the model development process [Betancourt \(2018\)](#) has proposed the *Principled Bayesian Workflow* which involves key decisions and model checks at critical stages of the model development process. Since then the workflow has been continuously refined and adapted to various use cases. For instance, [Gelman et al. \(2020\)](#) created a comprehensive summary of model validation techniques which is relevant to applied Bayesian statistics in general, while [Schad et al. \(2021\)](#) proposed a selected set of methods tailored towards cognitive sciences, including research on language. Specifically, [Schad et al. \(2021\)](#) identify four crucial validation steps in the model development process, including (1) prior predictive checks, (2) computational faithfulness, (3) model sensitivity, and (4) posterior predictive checks.

It is important to recognize that the principled Bayesian workflow is not a set of formal statistical tests but a collection of visual inspection techniques. These visualizations enable modelers to systematically identify and investigate potential issues in the model that may stem from their implementation choices.

This tutorial contributes to the ongoing refinement of the principled Bayesian workflow by highlighting and supplementing techniques that are particularly relevant for cognitive modelers working in the field of experimental semantics. That is, given the theory-rich foundation of semantics, we found it important to include a model validation step that ensures domain expertise (in the form of model priors) is appropriately incorporated into the computational model. Furthermore, due to the complexity of observational data in our field, the model should be capable of producing reasonable predictions. Therefore, we propose checking not only whether the model’s prior predictions generate sensible distributions of the outcome variable, but also whether predicted individual behavioral patterns align with theoretical expectations. In the same vein, we expand on the posterior predictive checks proposed by [Schad et al. \(2021\)](#). Rather than solely verifying if the population average is covered by the posterior distribution of the outcome variable, our approach recommends assessing the models capability to re-describe behavioral patterns at both the group and individual levels. Furthermore, we added additional checks to validate that the assigned prior distribution is the one produced by the sampling algorithm. As a result, our proposed principled Bayesian workflow addresses five *key questions* during the validation process:

1. Is the model consistent with domain expertise and capable of predicting theory-consistent behavioral patterns at the group and individual levels?
2. Is the computational algorithm (i.e., the model specification in Stan) implemented in a way that allows for accurate estimation of the posterior distributions?
3. Are the expected data and research design characteristics (e.g., number of participants, trials per participant, stimuli) sufficient to effectively update the prior distributions?
4. Is the model capable to recover the true data generating process?
5. Is the model capable of accurately redescrbing the key behavioral patterns in observed data at both the group and individual levels?

To equip researchers who apply the principled Bayesian workflow with the skills needed to adapt their models when challenges arise, the tutorial additionally includes a troubleshooting checklist for each model validation step. Please note that this tutorial assumes that the readers already have a basic knowledge of Bayesian inference, including the implementation of a Bayesian cognitive model in Stan and sampling from its posterior using Markov Chain Monte Carlo (MCMC) sampling. For an extensive introduction to Bayesian statistics, we recommend [Lambert \(2018\)](#) and [Wagenmakers and Matzke \(2023\)](#), for Bayesian hierarchical modeling we recommend [Veenman et al. \(2023\)](#) and [Rouder and Lu \(2005\)](#), for MCMC sampling [van Ravenzwaaij et al. \(2018\)](#); and for implementing MCMC with Stan, we point to the extensive online manual ([Stan Development Team, 2023](#)).

2 Case Study

The principled Bayesian workflow will be illustrated using the research of generalized quantifiers as a case study, with a computational model and data derived from a sentence verification paradigm, specifically a truth value judgment task. Variants of this experimental set-up are commonly discussed and have been used before to study the perception of meaning (among others, [Clark and Chase, 1972](#); [Szymanik and Zajenkowski, 2010](#); [Rück et al., 2021](#); [Spsychalska et al., 2016](#); [Scontras and Pearl, 2021](#)).

2.1 The Paradigm

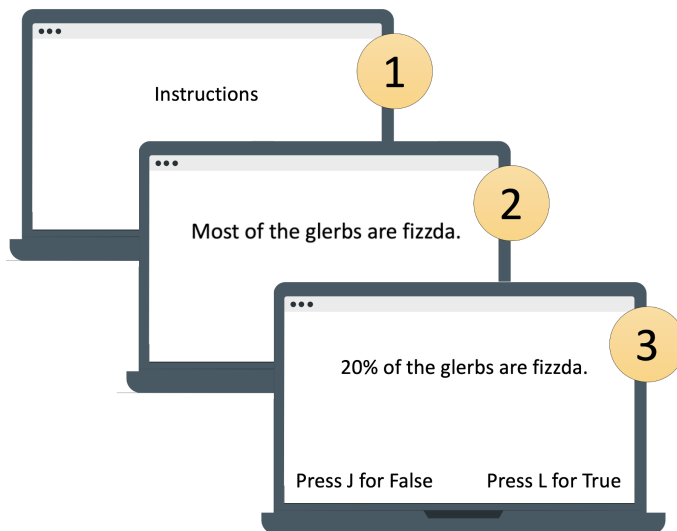


Figure 1: A single trial of a truth-value judgment task. First, instructions on the task are presented. Second, a sentence including the quantifier is presented, here *Most*. This is the part including the linguistic information. Third, a percentage is presented in a sentence with pseudo-words. Participants then have to choose whether this quantified sentence is true based on the linguistic information presented before.

The setup for a single trial in the truth-value judgment task, as implemented in [Ramotowska et al. \(2023\)](#), is illustrated in Figure 1. Participants were first instructed to read a quantified sentence containing pseudo-nouns and pseudo-adjectives (e.g., *Most of the glerbs are fizzda.*). Next, they read a second sentence in which the quantifier was replaced by a specific proportion (e.g., *20% of the glerbs are fizzda.*). Finally, participants were asked to judge the truth of the quantified sentence. The outcome variable in this task was participants’ response, which could be either ‘true’ or ‘false.’ This task featured pseudo-nouns and pseudo-adjectives instead of real words to eliminate any potential influence of prior knowledge about actual entities. Additionally, by presenting only proportions, quantifiers were judged based on a proportional reading ([Ramotowska et al., 2024](#)). Across trials, the authors varied both the presented percentages and the quantifiers. The percentages ranged from 1 to 99, excluding 50. The quantifiers used in the experiment included *Few*, *Many*, *Fewer than Half*, *More than Half*, and *Most*. For this tutorial, however, we will only consider the quantifiers *Few*, *More than Half*, and *Most*.

Figure 2a illustrates an ideal response pattern for a single participant in the truth-value judgment task for the quantifier *More than Half*. In this illustration, we assume that the participant adopts a threshold of 50%, such that for presented percentages below this threshold, the participant judges the quantified sentence as 'false,' while for percentages exceeding the threshold, the participant judges it as 'true'. An ideal responder makes these judgments consistently, which would result in a staircase pattern; for presented percentages below the meaning threshold the proportion of 'true' responses equal to zero while for presented percentages above the threshold the proportion of 'true' responses rise to one. Figure 2b illustrates a second response pattern for the same participant, this time for the quantifier *Most*. For this quantifier, the participant adopts the same meaning threshold; however, their threshold is associated with greater uncertainty, resulting in a larger vagueness around the threshold and a less steep response curve. The concept of vagueness provides semantic information, as it provides information about the probabilistic processes that lead to the response. This parameter should be distinguished from simple response noise, which may arise due to lapses in attention, accidental clicks, or misreading the sentences (Ramotowska et al., 2024). Figure 2c illustrates a potential response pattern for the quantifier *More than Half*, including a response error parameter that causes the entire response curve to be compressed.

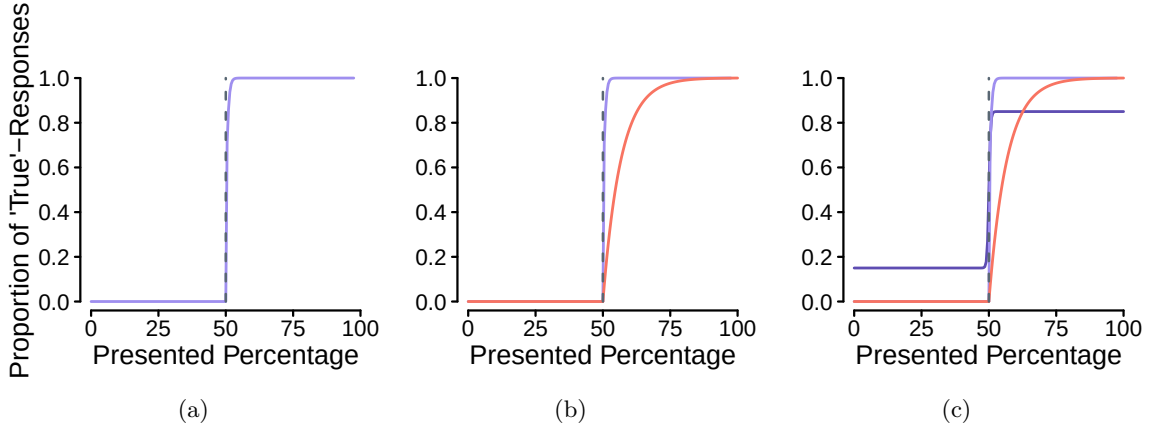


Figure 2: Illustration of the parameter structure with response patterns. The x-axes show the presented percentage in the task (i.e., the quantity in sentence 2 in each trial) on a scale between 0 and 100. The y-axes show the proportion of 'true'-responses of the participant. Panel a shows the pattern of an ideal (logical) responder for the quantifier *More than Half*. From a presented percentage of 50% on, the responder always answers with 'true'. Panel b adds a potential response pattern of a more uncertain responder and the quantifier *Most*. We can see uncertainty at values close to a presented percentage of 50% indicating vagueness. Panel c shows the response pattern for a noisy responder to the response pattern from Panel a. Below and above the threshold is still a chance to answer 'true' or 'false' due to an error (e.g., misclicking). Please note that the functions in this plot are only for illustration purposes and not based on real data.

2.2 Bayesian Cognitive Model

The Bayesian cognitive model we use in this tutorial was initially developed by Ramotowska et al. (2024) and further refined by Sarafoglou et al. (2024a). The cognitive model is a Bayesian three-parameter logistic regression model that captures semantic differences between quantifiers across three dimensions: meaning threshold, vagueness, and response noise. Additionally, the model is hierarchical, assuming that responses from an individual are drawn from a common population distribution.

Let i indicate individuals, j indicate the quantifier, k indicate each experimental trial. Then $Y_{i,j,k}$ is the i th participant's response to the j th quantifier in the k th trial. $Y_{i,j,k} = 1$ indicates that the participant answered 'true', $Y_{i,j,k} = 0$ that the participant answered 'false'. $Y_{i,j,k}$ is then modeled as a Bernoulli trial using an *exponential link* on the probabilities:

$$Y_{i,j,k} \sim \text{Bernoulli}(\pi_{i,j,k}),$$

$$\pi_{i,j,k} = \gamma_{i,j} + (1 - 2\gamma_{i,j}) \times \frac{1}{1 + \exp(-\mu_{i,j,k})},$$

where the second line maps the probability space of π onto the real space of μ . The parameter $\gamma_{i,j}$ represents the probability that individual i makes a response error for trials involving quantifier j . To ensure that participants do not perform worse than chance, $\gamma_{i,j}$ is truncated between 0 and 0.5. The remaining semantic components are encoded within $\mu_{i,j,k}$:

$$\mu_{i,j,k} = \begin{cases} \frac{c_{i,j,k} - \beta_{i,j}}{\alpha_{i,j}} & \text{if } c_{i,j,k} \geq \beta_{i,j} \\ 0 & \text{if } c_{i,j,k} < \beta_{i,j}. \end{cases}$$

where $c_{i,j,k}$ represents the presented percentage in a given trial (truncated between -0.5 and 0.5), $\beta_{i,j}$ denotes the meaning threshold (on the same scale as $c_{i,j,k}$), and $\alpha_{i,j}$ refers to the vagueness associated with each quantifier. Please note that whenever the presented percentage is lower than the meaning threshold, $\mu_{i,j,k}$ is equal to zero, meaning that an individual's response probability is determined solely by their response noise rate.

2.2.1 Model Description

Let us break down the equations and relate them to the paradigm. We model the response to each trial in the truth-value judgment task as a probability, $\pi_{i,j,k}$. The Bernoulli distribution captures the binary nature of the responses in the task: $0 = \text{'false'}$ or $1 = \text{'true'}$. The parameter $\gamma_{i,j}$ represents the response error rate, as introduced earlier. This parameter is constrained between 0 and 0.5, as higher error rates are implausible (individuals would perform worse than chance), and negative values are not meaningful.

Now we come to the core of the semantic representation in our model: $\mu_{i,j,k}$. This parameter represents the internal value that the individual assigns to a particular quantified sentence. But how do we derive it? $c_{i,j,k}$ serves as a fixed value—it represents the centered proportion based on the presented percentage in the task and is not estimated. To increase the robustness of the computational algorithm, this value is centered around 0 and scaled between -0.5 and 0.5 (e.g., -0.2 corresponds to a presented proportion in the task of 30%, and 0.1 refers to a presented proportion of 60%).

β_{ij} represents the meaning threshold for an individual with respect to a given quantifier. In this equation, the meaning threshold can be thought of as the necessary and sufficient condition for a quantified sentence to be considered 'true.' Only when the presented percentage exceeds this threshold does the probability of responding 'true' increase beyond the response noise rate. The meaning threshold parameter is represented on the same scale as $c_{i,j,k}$; that is, it is centered at 0 and ranges from -0.5 to 0.5 . The vagueness parameter $\alpha_{i,j}$ represents the uncertainty around the meaning threshold, with larger values leading to a less steep response curve. In our computational model, vagueness is represented on the log scale $\log(\alpha_{i,j})$, with bounds between -6 and 0 . We chose this transformation to assign the parameter a normal distribution as prior. The values of $\log(\alpha_{i,j})$ are then transformed back through exponentiation. Since the exponential of any real number is positive, this transformation ensures that α_{ij} remains within a positive range.

If the centered percentage falls below the threshold, $\mu_{i,j,k}$ will always be zero, meaning the individual's responses are solely driven by response noise. This model generates the asymmetric response pattern shown in Figure 2c.

Finally, it should be noted that the implementation of the cognitive model can only describe monotonic increasing response patterns. We observe these patterns for upward monotone quantifiers such as *More than Half* and *Many*. Quantifiers that result in monotonic decreasing response patterns, so-called downward monotone quantifiers, such as *Few*, cannot be described with the current implementation. To fit the model to experimental data from downward monotone quantifiers, we reverse the data before fitting the model. After the model has been fitted, the estimates are flipped back to match the original scale. We will mention this process again in the tutorial, as needed.

2.2.2 Prior Distributions

In a Bayesian hierarchical model, we assume that responses from individuals are drawn from a common population distribution. This means that when we assign prior distributions to the model parameters, we also need to specify the population distribution. We do this by placing additional prior distributions on the mean μ and the standard deviations σ of the model parameters:

$$\begin{aligned}\mu_{\log(\alpha),j} &\sim \mathcal{N}(-3, 1) \\ \mu_{\beta,j} &\sim \mathcal{N}(\mu_{\beta}, 0.1) \\ \mu_{\gamma,j} &\sim \mathcal{N}(0.17, 0.05)\end{aligned}$$

$$\begin{aligned}\sigma_{\log(\alpha),j} &\sim \text{Inv-Gamma}(2, 1) \\ \sigma_{\beta,j} &\sim \text{Inv-Gamma}(6, 1) \\ \sigma_{\gamma,j} &\sim \text{Inv-Gamma}(12, 1)\end{aligned}$$

The shape of each distribution is based on domain knowledge and calibrations informed by the model validation process conducted by [Sarafoglou et al. \(2024a\)](#). The value $\mu_{\beta,j}$, that represents the population mean for the meaning threshold parameter, varies across quantifiers: for *Few*, $\mu_{\beta} = -0.2$; for *More Than Half*, $\mu_{\beta} = 0$; and for *Most*, $\mu_{\beta} = 0.2$. After specifying the prior distributions at the population level, we now turn to the individual-level prior distributions, the shape of which are entirely determined by the population level:

$$\begin{aligned}\log(\alpha_{i,j}) &\sim \mathcal{N}(\mu_{\log(\alpha),j}, \sigma_{\log(\alpha),j}) \\ \beta_{i,j} &\sim \mathcal{N}(\mu_{\beta,j}, \sigma_{\beta,j}) \\ \gamma_{i,j} &\sim \mathcal{N}(\mu_{\gamma,j}, \sigma_{\gamma,j})\end{aligned}$$

3 Tutorial - Methods of the Validation Process

The following sections outline each step of the model validation process in sequence. We recommend following the order presented in this tutorial, as the steps build on one another. Given that the validation process involves extensive coding, which can be prone to errors, a good strategy is to cross-check the code with a colleague.

Please note that the methods for assessing prior predictives (step 1), computational faithfulness (step 2), model sensitivity (step 3), and parameter recovery (step 4) are applied to only one quantifier, as the models differ only in the prior distribution’s mean for the meaning threshold parameter. As such, these steps do not need to be repeated for all quantifiers. However, the posterior predictive checks (step 5) should be applied to all quantifiers, as the prior mean of the meaning threshold parameter is relevant to the results.

3.1 Prior Predictive Checks

We now address the first question in the validation process, namely whether the model is consistent with domain expertise and capable of predicting theory-consistent behavioral patterns at both the group and individual levels.

3.1.1 The Method

In this section, we focus on the prior distributions defined for our model. We recommend to assign prior distributions on model parameters that are plausible based on ones own expectations and the existing literature ([Lee and Vanpaemel, 2018](#); [Schad et al., 2021](#); [Aczel et al., 2020](#); [Lee and Wagenmakers, 2014](#)). However, prior distributions, for instance, on standard deviations or covariance matrices, often lack a natural interpretation, which makes it challenging to understand their impact on model predictions.

The first step in validating the model priors is to ensure that the sampling process is accurate. This can be done by drawing samples from each prior distribution in the model and plotting them as a histogram. Then, we can overlay the density of the distribution we assigned to each parameter on the histogram, which allows us to confirm that the sampled values match the theoretical prior distribution. Once the sampling process has been validated, we can proceed to analyze the prediction of the model.

We can study the impact of prior distributions on model predictions by simulating responses from hypothetical participants based solely on the prior distributions. These simulated datasets are commonly referred to as prior predictives. Figure 3 illustrates this process with four simulated datasets, each containing three participants. To generate a single simulated dataset, we first draw a sample for each prior distribution specified in our model (step 1 in Figure 3). Using this set of parameter values, we then simulate one dataset (step 2). This process is repeated as many times as needed to produce the desired number of datasets. Finally, the simulated responses are visualized both at the population level and at the individual level.

To verify whether the response data that are generated from the model are consistent with our theoretical expectations, it is necessary that the modeler formulates key characteristics in the data that need to be present. Essentially, we are asking “Given sets of parameter values drawn from our prior distributions, what kind of response data might we observe?”

In this case study, we aim for response data that resembles the one in Figure 2. The key characteristics of this data are as follows. First, we expect the proportion of ‘true’ responses to increase monotonically for upwards-entailing quantifiers and decrease monotonically for downwards-entailing quantifiers. This reflects the assumption that quantifiers have a single meaning threshold. Second, we expect the increase in the proportion of ‘true’ responses to occur around the presented percentages, which correspond to the center of our prior distribution for the meaning threshold, e.g., 70% for *Most*. Lastly, we expect response patterns to be indicative of low response noise, with the proportion of ‘true’ responses exceeding 70% for most individuals. It should be rare, if not impossible, for participants to perform at chance level, which would be reflected in a completely flat response pattern.

3.1.2 Application

Chunk 1 in the RMarkdown sets up the script, and *Chunk 2* contains the first Stan model. We structured this Stan model such that it draws samples from the prior distributions and simulates data based on these samples. In the model file, we define the parameters used in the model and specify their prior distributions. To simulate responses from hypothetical participants, we then include our model equations in the *generated quantities block*. Finally, we compile the model in R using the *rstan* package (see *Chunk 3*) (Stan Development Team, 2024), which rewrites the model code in a way that the *rstan* function can use it for sampling. We set-up the data structure for the Stan object (*Chunk 4*). This requires deciding how many hypothetical participants we want to include in our simulated data. We chose 43, as this matches the number of participants from whom we collected data in the pilot study. In addition, we specify the number of experimental trials each hypothetical participant completes. To match the real experiment, we chose 40 trials per quantifier, resulting in 120 trials per participant and a total of 5160 trials for the entire sample. As in the real experimental design, the presented percentages during the task are uniformly distributed across the 120 trials, with values ranging from 1 to 99, excluding 50. We decided to run two Markov Chain Monte Carlo (MCMC) chains, each consisting of 2,000 iterations. Of these, the first 500 iterations were used as a warm-up. The remaining 1,500 iterations provided 1,500 sets of samples, which resulted in 1,500 simulated data sets per chain.

Now we can begin the sampling process (*Chunk 5*). Since the model has not yet been fitted to data, this process will not be computationally intensive, although it will still take a few minutes. After sampling, we first extract the sets of samples from the joint prior distribution, along with the prior predictives (*Chunks 6-8*). To verify that the computational algorithm sampled correctly from our specified priors, we first plot the prior samples for each parameter in a histogram and overlay the theoretical density (*Chunks 9-10*).

Figure 4 shows the samples drawn from the prior distribution for the six parameters at the population level (i.e., the priors for the means and standard deviations of the three model parameters). Theoretical densities are overlaid with a solid line. We can see that the desired distributions are generally reflected in the samples for the parameters. However, there are some deviations: the means of the μ_β and μ_γ parameters are slightly shifted, and the three inverse gamma distributions (bottom panels) show fewer samples in the tails of the distributions.

This is *Problem A* from the troubleshooting checklist of this section (see below), and it is something we should investigate. Since coding errors, an insufficient number of samples, and convergence issues

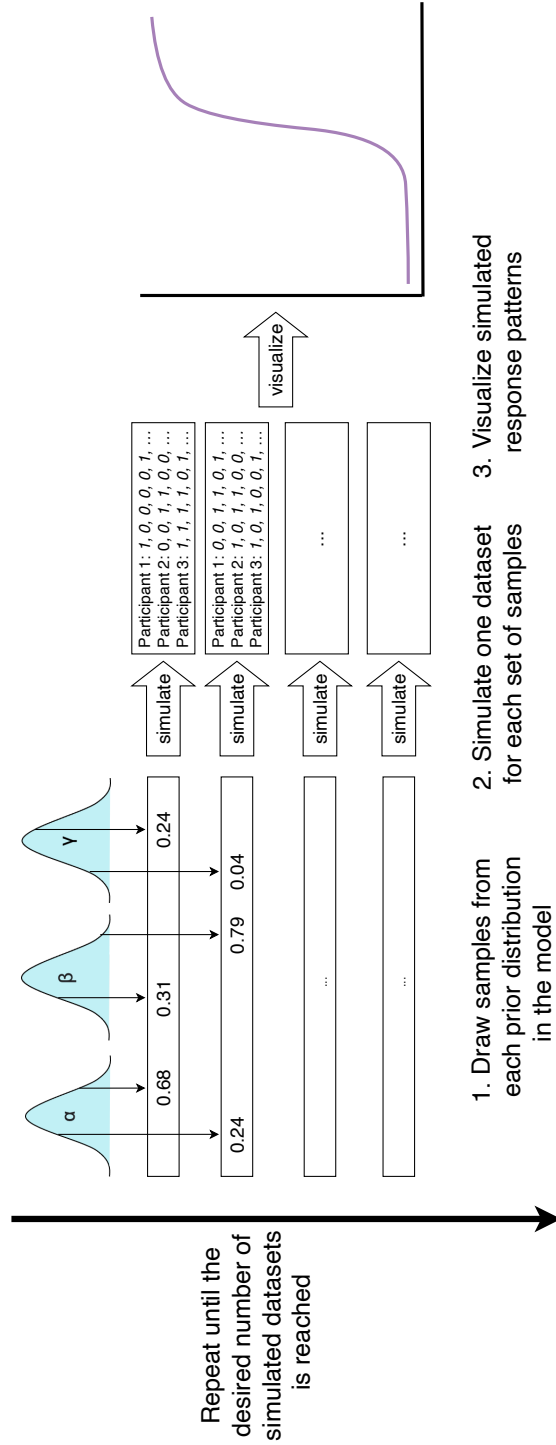


Figure 3: The process of prior predictive checks is illustrated using four simulated datasets, each featuring three participants. First, a value is drawn from each prior distribution specified in the model. Next, data are simulated based on these sampled values. Finally, the simulated data are visualized by aggregating across datasets and plotting the resulting response patterns. Our model passed the first validation check, if these response patterns align with theoretical expectations.

have been ruled out, we turn to *Solution A4*. In general, when the prior distributions specified in the model do not match the sampled distribution, the modeler must decide to what extent the priors can be modified and still remain aligned with the theory.

For instance, in [Sarafoglou et al. \(2024a\)](#), the authors observed a shift in their response noise parameter, μ_γ . They argued that it was implausible to assume that participants’ true response noise would be worse than chance, thus justifying the range restrictions on this parameter. However, the center of the prior distribution they assigned to the response noise distribution lacked a strong theoretical justification. Based on previous literature, the authors suggested that individuals generally exhibit low response noise rates, but there is no consensus on the exact value around which the response noise rate should be centered. Instead of relaxing restrictions, the authors therefore chose to slightly shift the center of the prior distributions to reduce the mismatch. If prior distributions cannot be adjusted for theoretical reasons or the mismatch cannot be fully eliminated, it is important to address the mismatch in the manuscript and account for it when using these priors to test predictions.

Figure 5 shows the correspondence between the samples and theoretical prior distributions after relaxing the numerical bounds on parameters (e.g., removing the fixed $[0,0.5]$ constraint on the guessing parameter). The sampled parameters now align more closely with their intended theoretical distributions. While we still ensure that predicted probabilities remain within $[0,1]$, we no longer enforce these constraints directly on the parameter values themselves. As a result, the mismatch observed under the previous constraints has been effectively eliminated. In the validation steps following this section, we will use the original simulated datasets, which are based on the samples with a slight mismatch. However, if modelers choose to proceed with *Solution A4* and modify the parameter priors to resolve the mismatch, the validation steps should be applied to the simulated data sets from the adjusted model.

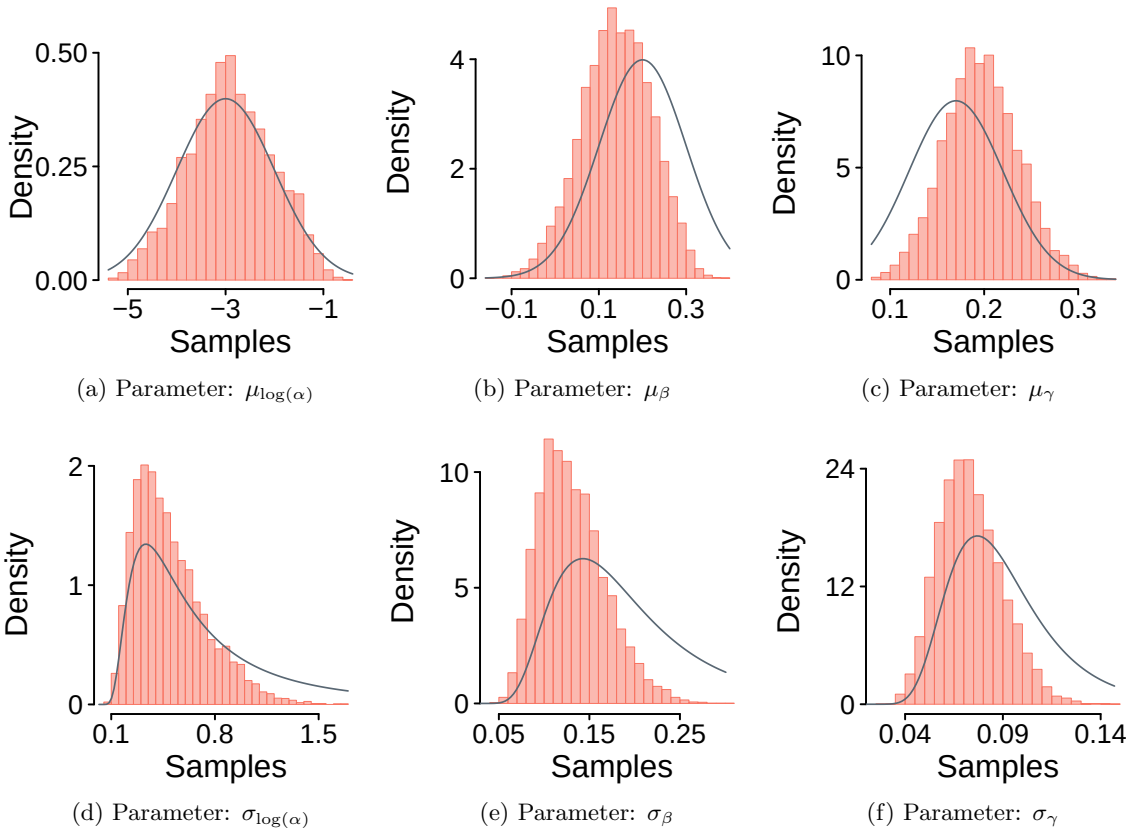


Figure 4: The samples from the prior distributions plotted as a distribution for the six hyper level parameters. We can see that the samples (bars) mostly reflect the underlying theoretical prior distribution (curve), however, some do not match perfectly, which requires further investigation.

We now turn to the prior predictive checks (*Chunks 11 to 13*). To evaluate the prior predictions

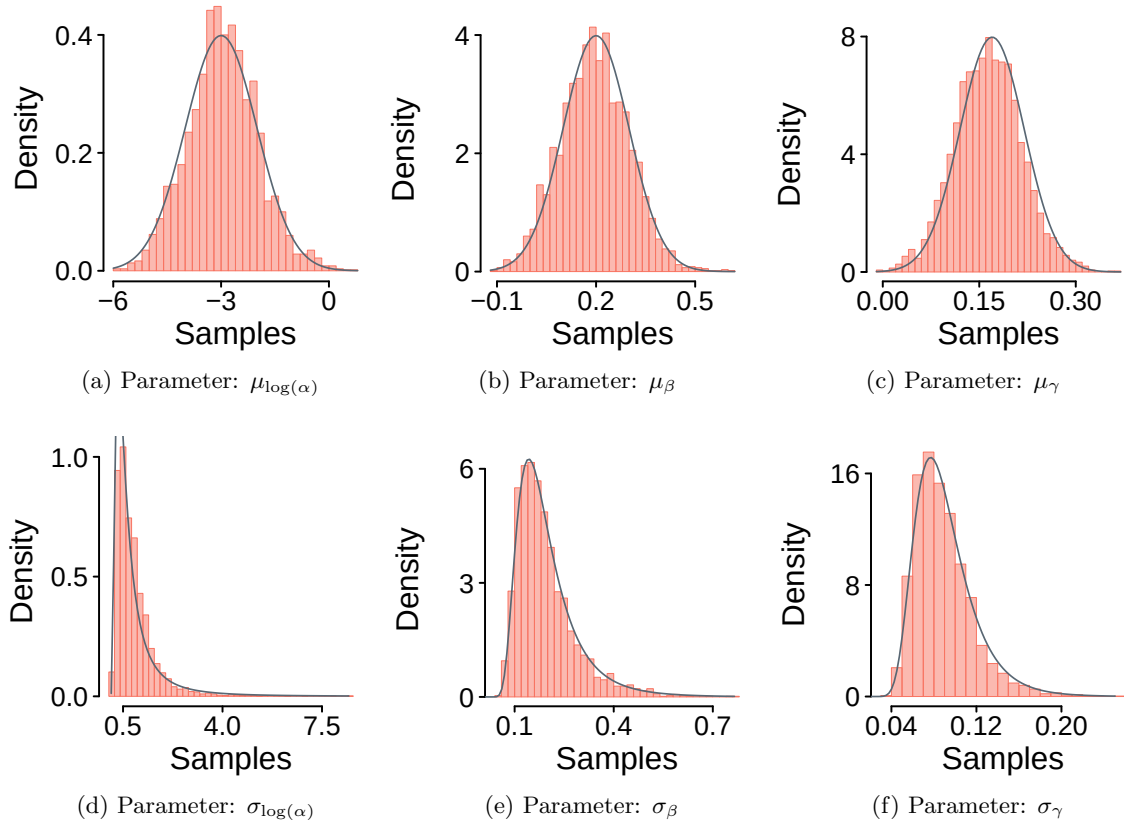


Figure 5: The samples from the prior distributions plotted as a distribution for the six hyper level parameters without truncation on the parameters. We can see that the samples (bars) almost perfectly reflect the underlying theoretical prior distribution (curve).

at the population level, we first averaged across all participants and all datasets, resulting in a single smooth curve that represents the mean predicted proportion of "true" responses as a function of the presented percentage. Consider Figure 6; on the x -axis, we plot the presented percentage in the task, while on the y -axis, we plot the proportion of 'true' responses. The prior predictions at the population level is presented as the dark solid line. Further, we averaged across all participants within each generated data set in a subset of 250 data sets. These are the lighter lines in the figure. Each of these light lines is the mean response pattern of 43 hypothetical participants combined for that particular simulated dataset. By plotting such lines, we demonstrate the variation in predicted response patterns that arise solely from different random draws of parameters from the prior.

To further examine the model's predictions at the individual level, we focus on a single hypothetical participant (participant 1) across a random subset of 250 simulated datasets (Figure 7). On the x -axis, we again plot the stimulus percentage, however, we plot it in categories to make the responses (consisting of 0 and 1) better visible. The solid red line represents the mean proportion of "true" responses across these datasets for the given participant. This plot helps to clarify whether individual-level response patterns align with the theoretical expectations suggested by the priors.

3.1.3 Interpretation

Figures 6 and 7 illustrate the prior predictive distributions on the population level and the individual level. In both cases, the response patterns resemble the ones depicted in Figure 2. With prior predictive checks, we ask ourselves: "Given multiple sets of samples drawn from our prior distribution, what kind of data might we observe?". At both the population and individual levels, we observed patterns that align with our theoretical expectations. Specifically, we see that the function increases monotonically for the upward entailing quantifier *Most*. This suggests that as the presented percentage in the task increases, the probability that participants will answer 'true' also increases. We can further see that at the population level the patterns show the highest uncertainty approximately around the expected threshold (for *Most*, $\mu_\beta = 70\%$). At the individual level, we observe greater variation; however, the response pattern does not conflict with our expectations or suggest extreme or unreasonable response behavior. We can therefore assume that we have selected appropriate prior distributions, and that our model is consistent with domain expertise.

Figures 6 and 7 are just one example of how the prior predictive check can be performed. For this particular paradigm, plotting the proportion of 'true' responses against the presented percentages is informative, as it provides insight into an individual's meaning threshold, the uncertainty they assign to this threshold, and their response noise rate. However, for different research designs, different visualizations may be necessary to provide meaningful insight into the response behavior at both the population and individual levels. We recommend selecting visualizations that either offer insights into the distributional properties of outcome variables or into the semantic processes that are central to the research question.

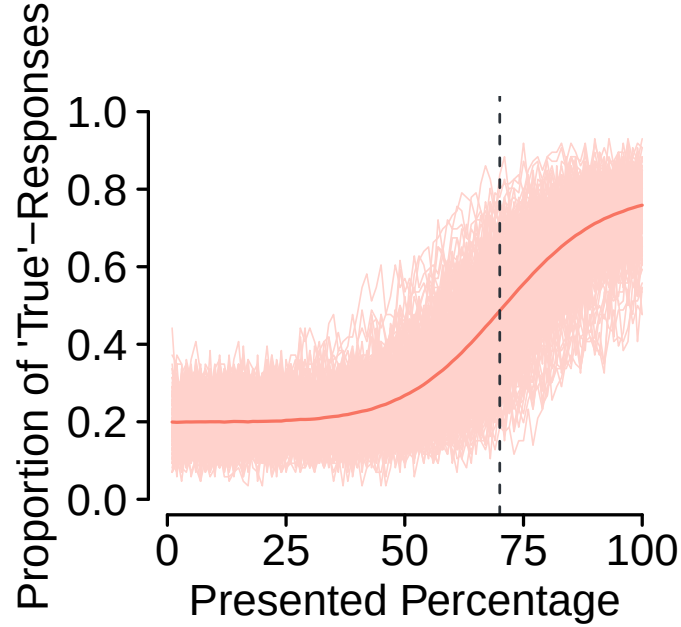


Figure 6: Aggregated prior predictives based on 250 draws from the joint prior distribution for the quantifier 'Most'. On the x -axis are the presented percentages in the task in %; on the y -axis are the proportion of *true*-responses. The function shows the desired pattern following the assumption of monotonicity and threshold in the expected area.

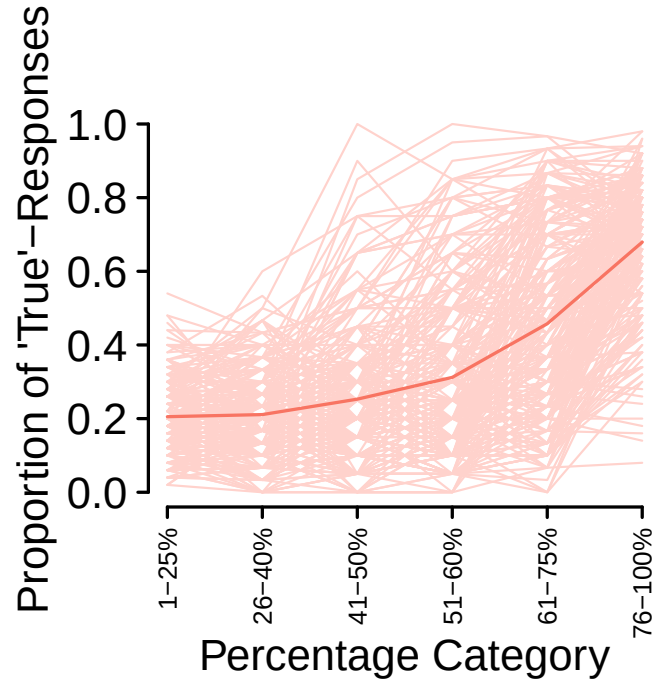


Figure 7: Predicted responses (light red lines) based on 250 draws from the joint prior distribution for a single hypothetical participant; and averaged responses across these data sets (solid red line) for the quantifier 'Most'. On the x -axis are the presented percentages in the task in % as categories; on the y -axis are the predicted responses and proportion of *true*-responses. The curve and the responses show the desired pattern following the assumption of monotonicity and threshold in the expected area.

Error Shooting Checklist

- Problem A: There is a mismatch between the samples drawn from the prior distributions (Figure 4) and the priors specified in the model.
 - Solution A1: As trivial as it may sound, this can always occur due to a coding error in the Stan model.
 - Solution A2: An insufficient number of samples may lead to response patterns or prior distributions that appear to deviate from what is expected. If in doubt, increase the number of samples to at least a few thousand to rule out this possibility.
 - Solution A3: Similarly, convergence issues can lead to unexpected response patterns or prior distributions. Therefore, perform MCMC diagnostics (e.g., plotting the chains and examining their autocorrelation) to ensure the chains have mixed properly.
 - Solution A4: Opaque priors can cause a mismatch between the assigned prior distributions and the prior samples. Specifically, in some models—especially those with heavy constraints—the specified prior distribution may not align with the distribution actually sampled (Merkle et al., 2023). If opaque priors are the source of the misfit, modelers can address the issue by relaxing the constraints, shift the prior distributions, or choose to retain the mismatch if the restrictions represent a necessary part of the theory (Merkle et al., 2023; Sarafoglou et al., 2024a). Further handling and less common sources of opaque priors are discussed in Merkle et al. (2023).
- Problem B: The prior predictives show the pattern following theoretical expectations, however not in the range of expected values.
 - Solution B1: The parameters are likely restricted in a wrong range. This requires rechecking the restrictions of all parameters in the model.
- Problem C: The prior predictive pattern appears reasonable at the group level. However, at the individual level the predictives seem unexpected, unreasonable, or extreme.
 - Solution C1: This issue may arise when modelers assign wide and uninformative priors to the model parameters, particularly for model parameters that encode the variability among participants. While these priors may produce reasonable response patterns if responses are aggregated across participants, they can lead to highly informative and unrealistic predictions for individual responses (Sarafoglou et al., 2024b). At the individual level, such broad priors may cause subgroups of participants to exhibit extreme responses in opposite directions to satisfy the model’s assumption of high variability. To address this, it is advisable to use informative priors, particularly by constraining the priors on parameters such as standard deviations or covariance matrices, in order to ensure a homogeneous population.

Take Home Message

Prior Predictive Checks. We ask whether the model produces predictions consistent with domain expertise. To assess this, we first plot the samples from the prior distributions of all parameters and check if they align with the priors specified in the model. Next, we simulate several datasets based on the prior samples, aggregate the responses across all datasets, and visualize them. Finally, we evaluate whether the predicted data patterns are reasonable and align with our expectations.

3.2 Computational Faithfulness

The next question in the validation process is whether the computational algorithm (i.e., the model specification in Stan) is implemented in a way that allows for accurate estimation of the posterior distributions.

3.2.1 The Method

Talts et al. (2018) introduced *simulation-based calibration* (SBC), a procedure to detect biases in computational algorithms. This validation step focuses not on how we formalize our expectations, but rather on the performance of the Markov Chain Monte Carlo (MCMC) algorithm used to approximate the posterior distribution. Specifically, in this step we will examine the draws from the posterior distribution which can be biased in two ways. First, posterior draws may fail to capture the true uncertainty of the model parameter, either by underestimating or overestimating its variance. Second, posterior draws could systematically deviate from the true posterior distribution by being consistently lower or higher (Talts et al., 2018; Schad et al., 2021; Kucharskỳ et al., 2021; Cook et al., 2006).

In the previous validation step, we generated simulated data from 250 sets of samples drawn from the prior distribution. For these simulated data, the specified prior distributions represent the true, known data-generating process. Simulation-based calibration relies on a key characteristic of every model: *self-consistency*. Self-consistency means that if we repeatedly draw samples from the prior distribution of a computational model, simulate datasets based on those sets of samples, and fit the same computational model to these datasets again, the posteriors should resemble the prior distributions, as no additional data is added. This is the crucial point, as the simulated data contains no additional information beyond what is specified in the prior. Empirical observations are not considered in this validation step. To pass this validation check, the resulting posterior distributions should then closely match the prior distribution (Talts et al., 2018; Schad et al., 2021).

A simple way to assess the self-consistency of a computational model would be to compare the mean or median of a large set of posterior samples with the prior distribution for each parameter. Unfortunately, this approach is not diagnostic, as it summarizes the distribution with single values. In contrast, simulation-based calibration can reveal not only whether the central tendency of the posterior distribution matches the prior expectation, but also whether the entire shape of the posterior distribution aligns with what the prior would produce.

The process of simulation-based calibration is illustrated in a path diagram in Figure 8. Similar to prior predictive checks, simulation-based calibration begins by drawing sets of samples from the prior distribution specified in our computational model. We then use each set of prior samples to simulate datasets for hypothetical participants. In this scenario, the sets of prior samples represent the true data-generating process, with each simulated dataset based on a different set.²

Next, we fit our Bayesian cognitive model to each of the simulated datasets. After sampling the posterior distributions, we compare the posterior samples to the original set of prior samples used to generate the dataset. Here, Talts et al. (2018) introduce the rank statistic as a diagnostic tool. Specifically, for each iteration (set of prior samples, simulated dataset, posterior samples), we count how often the posterior samples are smaller (or larger, this choice is arbitrary) than the original set of prior samples. This comparison allows us to determine where the original set of prior samples falls within the posterior distribution and helps us to identify if the posterior estimates of the model systematically underestimate or overestimate the true values. Cook et al. (2006) and Talts et al. (2018) proved that if the inference process works as intended, the ranks of the true values within the posterior samples should be uniformly distributed. This can be visually assessed by plotting the rank statistics in a histogram. A uniform distribution of ranks indicates an unbiased algorithm, while deviations suggest potential calibration issues (see the troubleshooting checklist and Figure 9).

The x -axis of rank statistic histograms represents the range of possible rank values. For instance, if we draw 250 samples from the prior, simulate data for each sample, fit the model, and then draw 100 posterior samples, the prior value can be larger than at most 100 posterior draws. In this scenario,

²Note that there is no need to run a separate process; the samples and datasets from the prior predictive checks can also be used for simulation-based calibration.

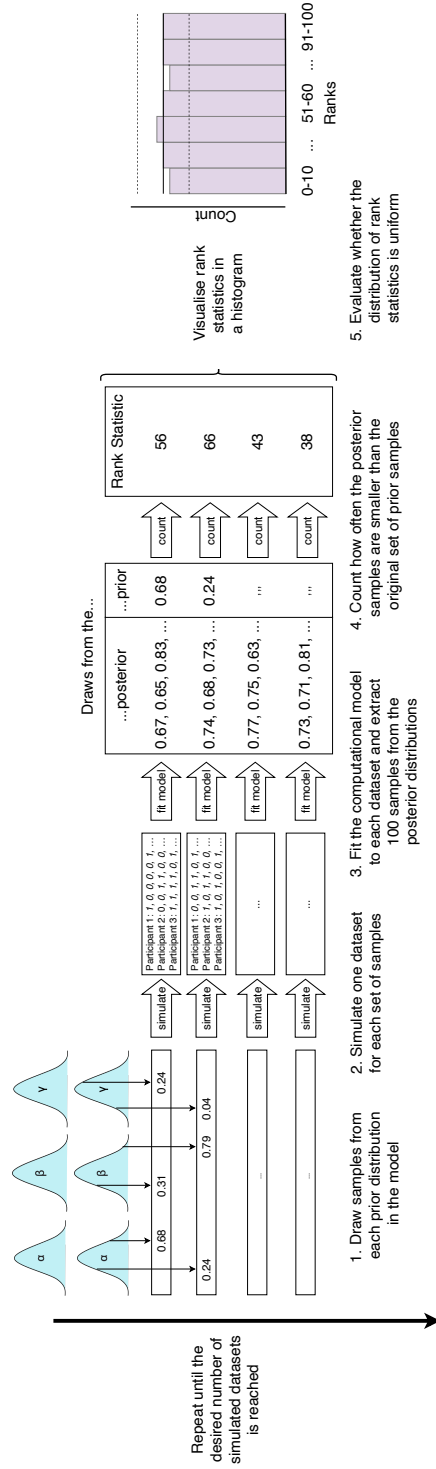


Figure 8: Process of Simulation-Based Calibration (SBC), illustrated here with four prior samples. The SBC process begins by drawing samples from the joint prior distribution, representing potential parameter values before any data has been observed. Using each of these prior samples, we then simulate synthetic datasets as if these values represented the “true” data-generating process. Next, we fit the Bayesian model to each of these simulated datasets. For each model fit, we draw posterior samples. We compare these posterior samples with the original prior sample used to generate the data for every set of prior samples. Specifically, we calculate how often the posterior samples are smaller (or larger, this choice is arbitrary) than the true data-generating value from the prior.

100 represents the maximum possible value of the rank statistic. Consequently, the x -axis spans from 0 to 100.

Figure 9 (adapted from Talts et al., 2018; Schad et al., 2021) illustrates the various patterns that may appear in the rank statistic for a given parameter. Figure 9a shows the pattern that we aim for. The posterior distribution closely resembles the prior, and as shown by the uniform distribution of the rank statistics.

Figure 9b and 9c illustrate emerging patterns when uncertainty is overestimated or underestimated, respectively. If the uncertainty is overestimated, there will be relatively few cases where the posterior draws are smaller than the prior value for most draws or for very few. This results in a $\cap$ -shaped distribution. Conversely, if the uncertainty is underestimated, there will be many cases where the posterior draws are either smaller than the prior value for most draws or for very few, leading to a U-shaped distribution. Finally, consider the case in which the algorithm is biased toward generating smaller posterior estimates compared to the true prior sample that produced the data. In such cases, for most generated datasets, the prior value will be larger most posterior draws. As a result, when counting how often the posterior draws are smaller than the prior value, we obtain large values for the rank statistics. Consequently, the corresponding distribution of rank statistics, when visualized in a histogram, will be right-skewed, as illustrated in Figure 9d.

To improve interpretability, we can add credible intervals to the histogram. While these credible intervals are not intended as a formal test, they serve as a helpful tool to identify the area that should be covered by the bins if the distribution were uniform. Since there are $L + 1$ possible ranks (from 0 to L), dividing them evenly into n_{bins} bins yields a uniform probability of $\frac{1}{n_{\text{bins}}}$ for each bin. Thus, the 95% credible interval is derived from a Binomial distribution $\text{Binomial}(N_{\text{prior}}, p)$, where N_{prior} is the number of prior samples, L the number of posterior samples, n_{bins} the number of bins, and $p = \frac{1}{n_{\text{bins}}}$. For example, if $n_{\text{bins}} = 10$, then $p = 0.1$. (Talts et al., 2018).

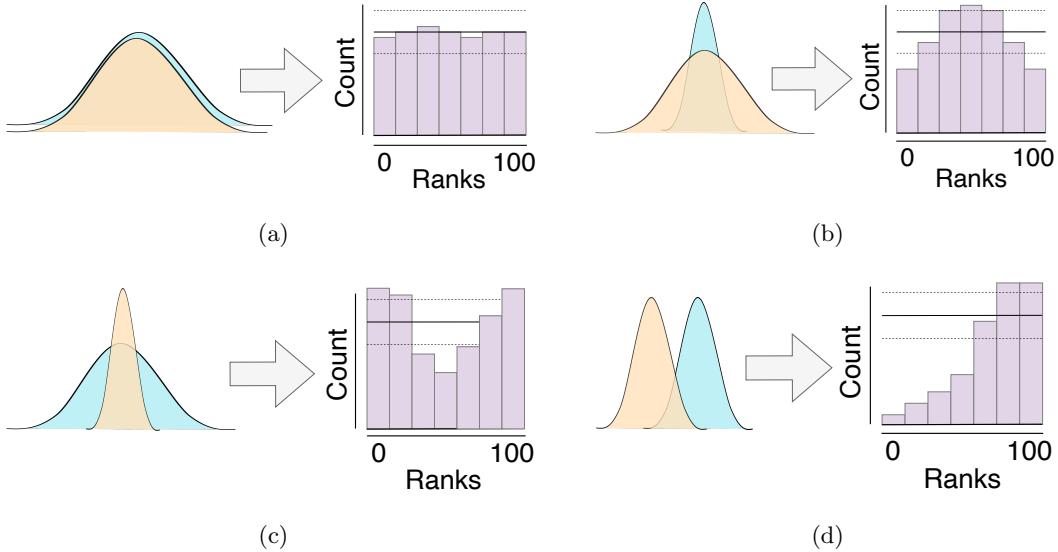


Figure 9: An example plot showing an ideal (a) and problematic (b-d) patterns in SBC rank statistic histograms and their underlying prior (blue) and the averaged posterior (orange) distributions. The pattern in panel b emerges when the variance is over-estimated, panel c when it is under-estimated. The pattern in panel d emerges when the algorithm is biased towards lower values. If the pattern in d would be in the opposite direction, it would indicate a bias towards higher values. Figure adapted from Talts et al. (2018); Schad et al. (2021).

3.2.2 Application

Since the model is now intended to fit the simulated data and generate samples from the posterior distributions, we adjusted the Stan code accordingly (*Chunk 14*). During the prior predictive checks,

we determined the number of sample sets from the prior distribution to be 250. We will now use these 250 samples from the prior distribution and the associated datasets again in the simulation-based calibration.

The number of datasets roughly determines how many samples to draw from the posterior distribution for computing our rank statistics and how many bins to use in the histogram. Generally, for this validation step, we use about half as many samples from the posterior distribution as we have prior values. With 250 prior values, we chose to draw 100 samples from the posterior distribution and used 10 bins to visualize the rank statistics in the histogram. Note that too many bins may distort the rank statistics distribution due to sampling variation, while too few bins may prevent us from identifying problematic patterns.³

We drew a total of 2,000 iterations from the posterior distribution. The first 500 iterations were used as a warm-up, and from the remaining 1,500 iterations, we retained every 15th sample to obtain the 100 posterior samples. *Chunk 15* specifies the number of posterior samples and the number of bins in our histogram and define a function to fit our computational model to the data. Now that everything is prepared, we can let the model fit to the 250 datasets (*Chunk 16*). This is the right moment to calmly get a coffee and enjoy the weather outside...

After the sampling we can create the rank statistic (*Chunk 18*), the count of posterior samples that are greater than the true value from the prior. Finally, we can plot the histograms with the function that is defined in *Chunk 19*.⁴

3.2.3 Interpretation

With simulation-based calibration we ask ourselves whether the computational algorithm is implemented in a way that allows for accurate estimation of the posterior distribution. Figure 10 presents histograms of rank statistics for all group-level parameters in our computational model, as well as for randomly drawn individual-level parameters, all for the quantifier *Most*. The rank statistic distribution should be approximately uniform if the algorithm is able to accurately recover the prior distribution.

While there are small deviations from uniformity in some of the figures (e.g., Figure 10a, Figure 10e), we do not observe any severe or systematic deviations. Overall, the rank statistic distribution is satisfactorily uniform for all parameters. Therefore, we conclude that our MCMC algorithm successfully recovered the prior distributions in the simulations.

³Ultimately, the number of simulated datasets and posterior samples depends on computational feasibility; complex models, such as the one presented here, may take a long time to fit to data. For less complex models, modelers can opt to use a higher number of simulated datasets and posterior samples for diagnostics. For example, in their application, [Schad et al. \(2021\)](#) used 1,000 sets of prior samples, 500 samples from the posterior distribution, and 20 bins in their histogram.

⁴For some interesting alternative visualizations, see for instance https://hyunjimoon.github.io/SBC/articles/rank_visualizations.html.

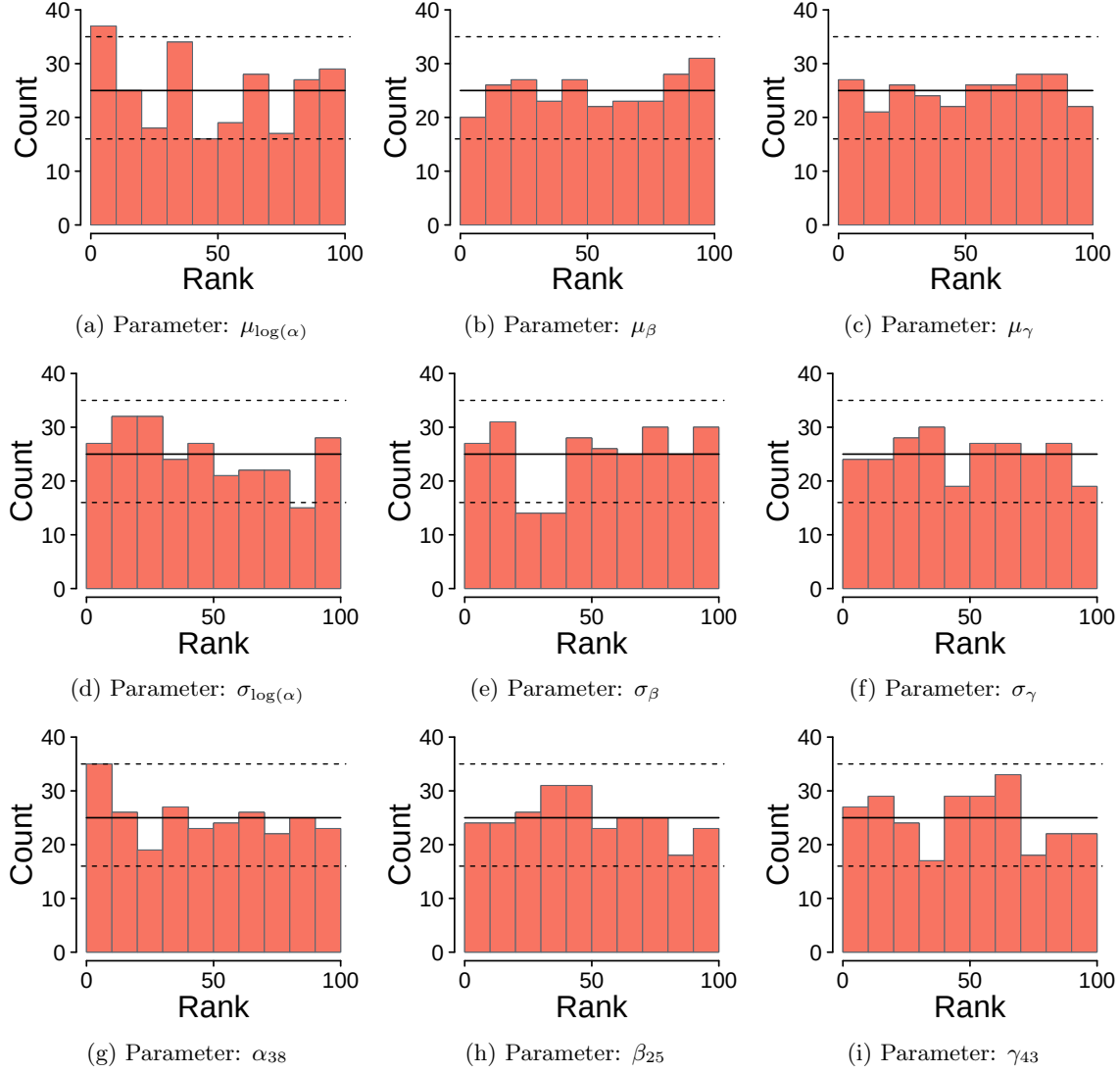


Figure 10: Histogram for the SBC rank statistic for the quantifier *Most*. Presented are the six hyper level parameters and three randomly drawn individual level parameters. The dashed lines illustrate the lower and upper limits of the 95% credible interval of a uniformly distributed variable. A perfectly uniform distribution would follow the black horizontal line. No distribution shows a problematic pattern and we can assume an unbiased algorithm.

Error Shooting Checklist

- Problem A: The histogram shows the problematic pattern Figure 9c.
 - Solution A1: If the histograms display a pattern similar to the U-shaped form in Figure 9c, or even more extreme, this may indicate strong autocorrelation in the posterior samples. To check for autocorrelation, use the `acf()` function in base R on posterior samples. If autocorrelation is present, chain thinning can help mitigate it. However, if the autocorrelation is too severe, reparameterization of the Stan model may be necessary (see, e.g., Papaspiliopoulos et al. (2007)). It should be noted that Stan has comparably low problems with auto correlation (see e.g., Annis et al. (2017)).
- Problem B: The histogram shows problematic patterns in Figure 9b or Figure 9d.
 - Solution B1: Some problematic patterns may emerge due to convergence issues. In packages like rstan, convergence problems can be easily identified using built-in MCMC diagnostics. If the model experiences convergence issues, the modeler may choose to use non-central parameterization, which can improve the efficiency of the sampling algorithm (Talts et al., 2018; Papaspiliopoulos et al., 2007). In Bayesian hierarchical modeling, non-central parameterization expresses individual-level parameters as deviations from the group-level means.
- Problem C: The sampler for the simulation-based calibration begins running but gets stuck after fitting the model to the first simulated dataset.
 - Solution C1: During simulation-based calibration, the MCMC sampler stores a large amount of numerical data, which requires significant memory storage. Consider computing all statistics and quantities after each iteration and then deleting the large dataset of posterior samples.

A note on identifiability: Problematic patterns in the SBC might also be caused by identifiability issues; where different parameter values produce indistinguishable data distributions. While severe non-identifiability might show up for instance as overly broad posterior distributions in the SBC, it may not always produce a clear pattern. Identifiability problems can be diverse and depend on the model at hand, thus, we can not recommend a general solution to this. If you suspect identifiability problems (for instance based on the later Model Sensitivity analysis), consider reparameterizing your model, adding more informative priors, or simplifying the model structure to enhance parameter distinguishability.

Take Home Message

Computational Faithfulness. We ask whether the computational algorithm is sufficient to accurately estimate the posterior distributions of the model parameters. To assess computational faithfulness, we verify the self-consistency of our model. We simulate datasets based on sets of samples from the prior distributions, fit our computational model to each dataset, and compute how many posterior samples are smaller than the prior value that generated the dataset. We then visualize these rank statistics to identify any problematic patterns.

3.3 Model Sensitivity

We now address the third question in the validation process, that is, whether the expected data and research design characteristics (e.g., number of participants, number of trials) are sufficient to effectively update the prior distributions.

3.3.1 The Method

In this section, we assess *model sensitivity*, which focuses on how effectively the expected data inform the posterior distributions. As proposed by [Schad et al. \(2021\)](#), we compute the model sensitivity using two quantities: posterior z-scores and posterior contraction estimates. To evaluate a model’s sensitivity, we can use the results from the simulations conducted during the simulation-based calibration in the previous section. Specifically, we use the sets of prior samples and their corresponding datasets to compute both quantities.

Posterior z-scores are useful to assess whether a posterior distribution is close to the original value that generated the data and sufficiently certain. These scores are obtained by first computing the difference between the posterior mean and the original parameter values and then scale the result by the standard deviation of the posterior distribution (i.e., the posterior uncertainty). The resulting z-score indicates how closely the entire posterior distribution aligns with the true parameter value:

$$z = \frac{\mu_{\text{post}} - \mu_{\text{true}}}{\sigma_{\text{post}}},$$

where μ_{post} refers to the posterior mean, μ_{true} the original parameter values, and σ_{post} the standard deviation of the posterior distribution. Z-scores close to 0 are desirable, as they indicate an optimal balance between the difference between the posterior mean and the true value relative to the posterior variance. Therefore, the z-score is well suited to describe parameters on different scales. A smaller posterior variance requires a smaller difference between the posterior mean and the true value to achieve a z-score near 0. Conversely, if the posterior variance is high, the difference between the posterior mean and the true value can be greater, while still achieving a z-score around 0. While there are no formal guidelines, [Schad et al. \(2021\)](#) suggest that z-values greater than 4 or less than -4 should occur only rarely, and values exceeding these thresholds may indicate problematic patterns.

The second key quantity are posterior contraction estimates. These estimates quantify the reduction in uncertainty from the prior to the posterior distribution. This approach is intuitively reasonable: when the model encounters data, the uncertainty of the prior should decrease, resulting in a lower posterior variance:

$$s = \frac{\sigma_{\text{true}}^2 - \sigma_{\text{post}}^2}{\sigma_{\text{true}}^2},$$

where σ_{post}^2 refers to the variance of the posterior distribution and σ_{true}^2 to the variance of the prior distribution. The difference between the variances in the numerator mainly determines the contraction estimate. When the posterior variance is small, the numerator will be close to the denominator, resulting in contraction estimates near 1. This is the desirable result. If the posterior variance has not been effectively updated and remains close to the prior variance, the contraction estimate will be close to zero. In rare cases, if the posterior variance exceeds the prior variance, the contraction estimate may be less than zero. These cases are less related to our expected data and may instead indicate convergence issues or identifiability problems in the computational model.

The two key quantities are then visualized in a single figure. Based on a figure presented by [Schad et al. \(2021\)](#), Figure 11 illustrates both the ideal patterns of z-scores and contraction estimates, as well as problematic ones.

The ideal pattern, presented in area A, is one where the posterior z-scores are concentrated around 0, and the posterior contraction estimates approach 1. In this case our expected data contribute to the recovery of the original data generating value and to an effective updating of the posterior distribution.

The pattern in area B displays the case in which there is little change from the prior to the posterior distribution. In this case, the posterior means will reflect the original values (as those were drawn from

the prior distribution), but the contraction estimates would cluster around 0 as the uncertainty was not sufficiently reduced. This pattern can occur in models with identification problems, which may arise from overly complex models (i.e., models with too many parameters) or from data that lack sufficient information or variability. In both cases, the data are unable to sufficiently inform all parameters. A similar pattern can occur if the prior is too informative (i.e., it strongly constrains the parameter values), which may result in estimates reflecting the prior more than the data—especially when data are sparse. Conversely, if the prior is too vague, this can exacerbate non-identifiability, leading to unstable or overly uncertain estimates. Both cases highlight the importance of balancing the prior’s informativeness with the available data.

If the expected data are unable to effectively update the prior distribution and the original values that generated the data cannot be recovered, we observe a pattern similar to the one displayed in area C. When the computational model is fit to data generated from prior samples, concentration in area C should not occur. However, this pattern may arise when models are fit to real data (e.g., from a pilot study) and can reveal a prior/likelihood conflict, combined with one of the issues that also contribute to the patterns observed in area B. In a prior/likelihood conflict, the prior may suggest a specific range or distribution for a parameter, while the data (through the likelihood) suggest a very different range, which can lead to biased posterior estimates.

Finally, expected data may update the posterior distribution but still lead to biased posterior estimates, resulting in patterns displayed in area D. This can have two causes. First, it may be due to a prior/likelihood conflict, which pushes the estimates into incorrect ranges of the distribution. Second, the computational model may overfit the data. Overfitting occurs when the model mistakenly interprets noise in the data as part of the model parameters. In this case, the noisy data reduce the uncertainty of the posterior distribution, but the model systematically over- or underestimates the original values that generated the data.

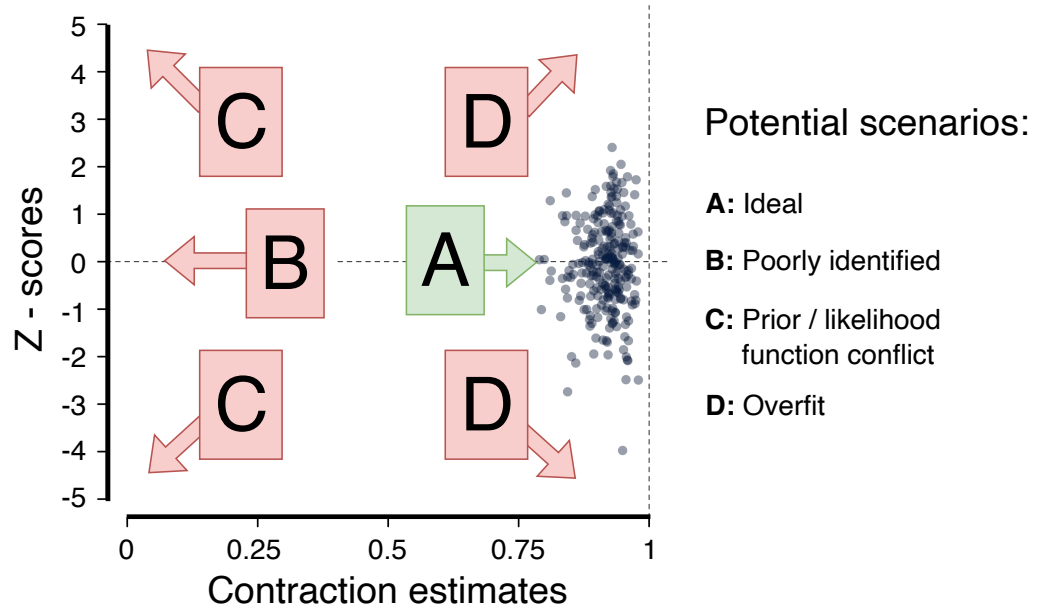


Figure 11: Posterior z-scores and posterior contraction estimates in a scatter plot (Figure adopted from [Schad et al. \(2021\)](#)). z-scores around 0 are ideal; contraction estimates around 1 are ideal. Thus, a concentration presented in this plot is desirable. Concentration in other areas of the diagram (here marked with the arrows), may indicate specific problems like overfitting or poor identifiability of the model.

3.3.2 Application

During the simulation-based calibration, we saved the variances of all 250 posterior distributions for the model parameters, and the 250 sets of samples from the prior distribution that generated the data (the true values). To obtain each of the posterior z-scores, we scale the difference between the posterior mean and the original prior sample value by the standard deviation of the posterior distribution. To compute each of the posterior contraction estimate, we scale the difference in variance for each of the posterior distributions with the variance of the 250 samples from the prior distributions (one value for each parameter) again by this variance of the 250 samples from the prior distributions. These calculations are carried out for each model parameter (see *Chunk 20-21*). Following the recommendations of [Schad et al. \(2021\)](#), we set the y-axis range to cover posterior z-scores from -4 to 4 (*Chunk 22*).

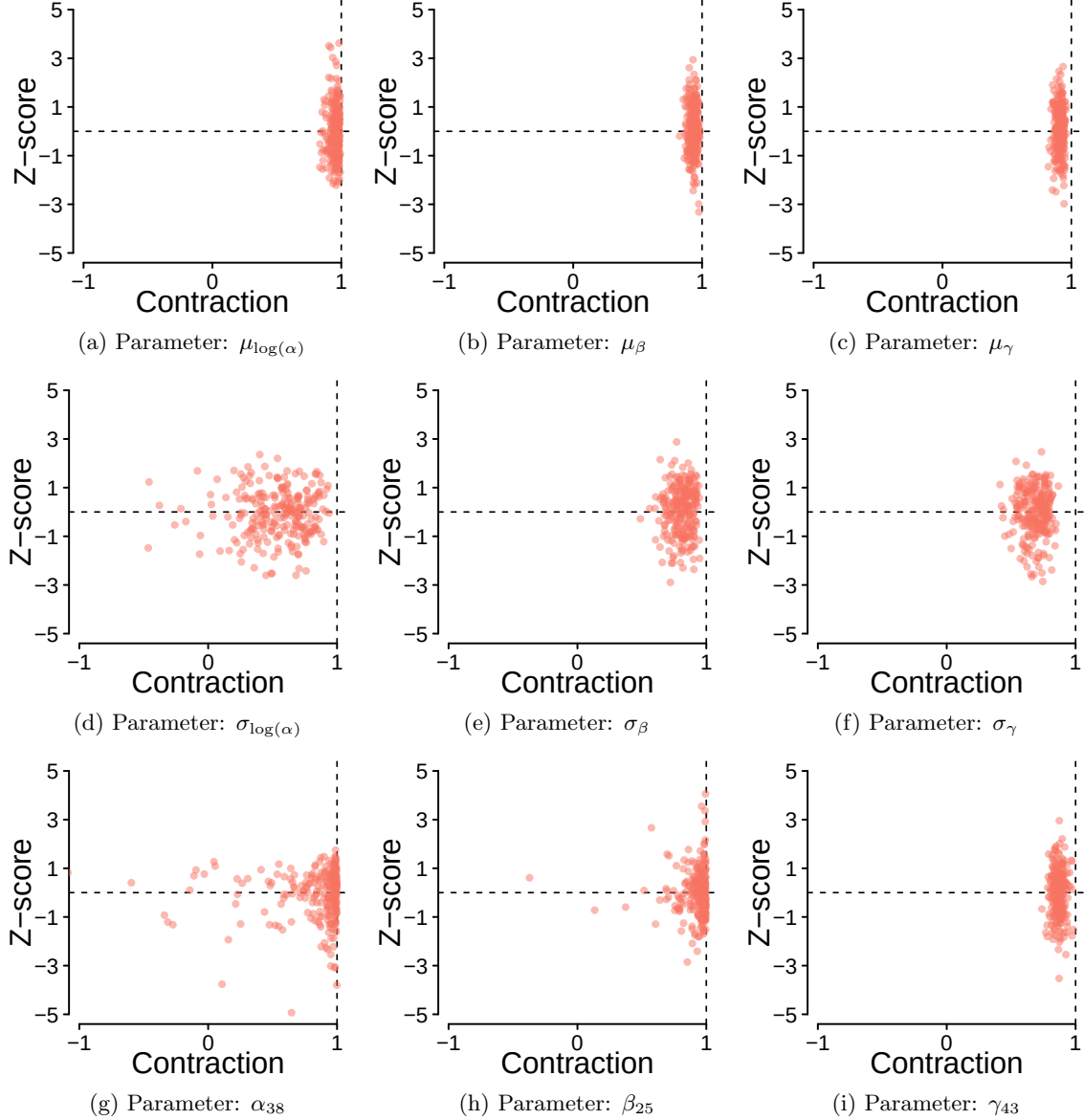


Figure 12: Posterior z-scores and posterior contraction estimates in a scatter plot for the quantifier *Most*. Presented are the six hyper parameters and three randomly drawn individual level parameters. We can see the desired pattern for all quantifiers. The vagueness σ parameter shows the most prevalent deviation, but still in an acceptable range.

3.3.3 Interpretation

We visualize the posterior z-scores and posterior contraction estimates for the group-level parameters and randomly drawn individual-level parameters in Figure 12 for the quantifier *Most*. As shown, they predominantly fall within the ideal area, with posterior z-scores around 0 and posterior contraction estimates around 1. Based on the visual inspection of the model sensitivity plots, we conclude that the updating process works as expected and that the data, including the number of participants and trials, provide sufficient information.

3.4 Parameter Recovery

In addition to assessing computational faithfulness and model sensitivity, we can evaluate the model's ability to accurately recover parameters by visualizing the estimated posterior means against the original values that generated the datasets in a scatterplot.

3.4.1 The Method

Knowing the true data-generating parameter values behind our prior predictives allows us to compute the correlation between the true value and the estimated posterior mean. This method offers a quick way to assess how well the estimated posterior distribution recovers the true parameters. A high correlation between the true value and the estimated posterior mean indicates good parameter recovery. While computational faithfulness and model sensitivity checks provide more detailed insights into the performance of the computational algorithm and the model, we recommend the parameter recovery plot as an additional useful, easy-to-implement diagnostic in the validation process.

3.4.2 Application

To assess parameter recovery, we can create scatter plots with the x -axis representing the original values that generate the simulated data and the y -axis showing the estimated values across the 250 iterations. This step is executed in *Chunk 23* in the supplied RMarkdown file.

3.4.3 Interpretation

Figure 13 illustrates the recovery of parameters for the quantifier *Most*, for all parameters at the group level of our computational model and randomly drawn parameters at the individual level. We also computed the Pearson correlation coefficient to aid us in interpretation. The values in the scatter plot should be close to the diagonal identity line drawn in the plots. We can see that the correlations are satisfactory across all parameters. Only parameter $a[38]$ shows a slightly atypical correlation pattern with values clustering at the lower ranges of both the estimated and true values, likely due to the log-transformation applied to this parameter. However, the true and estimated values still align, and the correlation remains sufficient ($\rho = 0.82$).

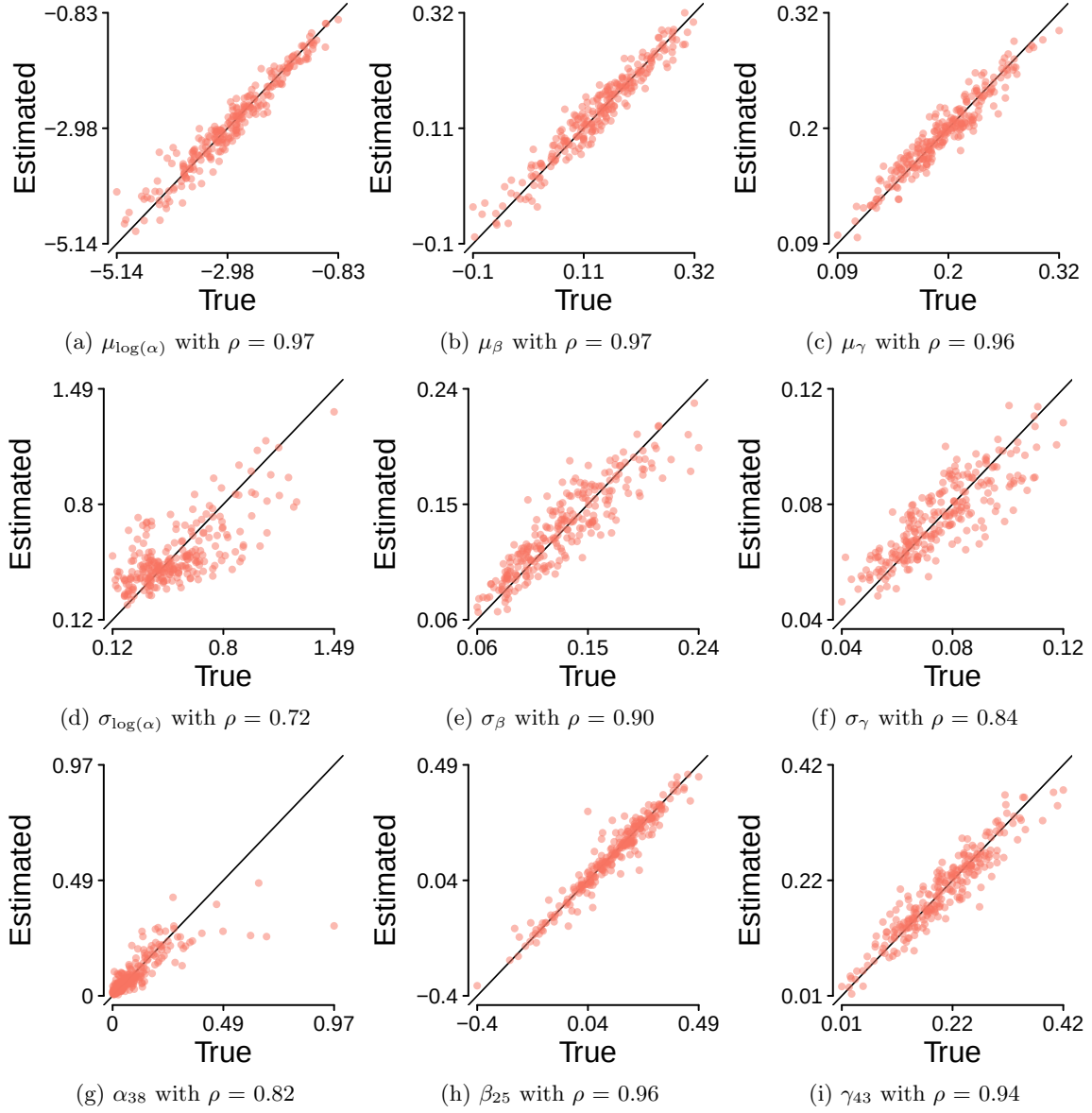


Figure 13: True vs. estimated parameter correlation for the quantifier *Most*. Presented are the six hyper parameters and three randomly drawn individual level parameters. We see a high correlation for all parameters and can therefore assume a satisfactory parameter recovery.

Error Shooting Checklist

- Problem A: The contraction vs. z-scores scatter plot shows problematic concentration in area B of Figure 11.
 - Solution A1: This pattern may arise if the data lack sufficient information to effectively update the estimates. To address this, the first step is to increase the number of hypothetical participants and trials in the simulated data (and, by extension, in the planned empirical study). Additionally, refining the model structure—such as simplifying the model or using less informative priors—can improve the estimation process.
- Problem B: The contraction vs. z-scores scatter plot shows problematic concentration in areas C or D of Figure 11.
 - Solution B1: When the computational model is fit to data generated from prior samples, concentration in area C should not occur. Any anomalies detected in such setups are indicate of issues in the code or estimation algorithm. When the computational model is fit to real data (e.g., from a pilot study), it may reveal a prior-likelihood conflict. In such cases, it is necessary to analyze how the data violate the constraints imposed by the prior distribution. If the data are considered typical and of high quality, modelers may address the conflict by ensuring that the range restrictions of the prior do not interfere with observed data patterns.
 - Solution B2: There is no single typical cause of overfitting. Reducing the complexity of the model can help prevent the model from overinterpreting the noise in the data. Additionally, visualizing the data associated with the highest z-scores may reveal problematic data patterns and issues in simulating realistic data from the priors.
- Problem C: The parameter recovery scatter plot shows a low or no correlation between the true and estimated values.
 - Solution C1: Similar to problem A, this may indicate an identifiability issue, where certain model parameters are not well-informed by the data. To address this, we can increase the informativeness of the simulated data (i.e., increasing the number of participants and trials), simplify the model, or assign more informative priors to the parameters.
 - Solution C2: Conversely, if the prior is too vague, this can exacerbate non-identifiability, leading to unstable or overly uncertain estimates.

Take Home Message

Model Sensitivity and Parameter Recovery. We ask whether the data provide sufficient information to effectively update our prior beliefs and address the research questions. In this model validation step, we visualize the true vs. estimated values to quickly assess the parameter recovery of the computational model. In addition, we examine two components of the posterior distribution: the posterior z-scores (which measure how well the estimated posterior mean matches the true parameter used to simulate the data) and the posterior contraction estimates (which quantify the reduction in uncertainty from the prior to the posterior). Visualizing these quantities helps identify potential issues related to the informativeness of the data, conflicts between the data and priors, or an overly complex model structure.

3.5 Posterior Predictive Checks

The final question for model validation is whether the model can accurately capture the key behavioral patterns observed in the data at both the group and individual levels.

3.5.1 The Method

Thus far, all model validation steps have been performed before fitting the model to empirical data. Posterior predictive checks are performed after the model has been deemed adequate and empirical data have been collected. Since the model is no longer subject to modification at this stage to avoid overfitting, posterior predictive checks serve as a sanity check to assess its predictive adequacy. They should not be used to guide further model adjustments unless the refined model is fitted to new data.

To perform posterior predictive checks we thus first fit the model to real data, and then simulating data based on the posterior parameter estimates. The simulated data are then compared to the observed data (Rubin, 1984; Gelman et al., 1996). There are several reasons why the pattern of simulated data may differ from the observed data. Although we have already examined the model’s fit after sampling, models are inherently abstractions of reality and may overlook critical aspects of the data-generating process.

Posterior predictive checks can reveal areas (e.g., in the tails of the distribution) where the model consistently performs poorly at predicting data (Schad et al., 2021). Moreover, data may vary in unexpected ways that conflict with the restrictions placed on the prior distributions (the so-called prior/likelihood conflict), which could indicate that the data are of insufficient quality, that participants may not have adhered to task instructions, or that the assumptions formalized in the model need to be refined in future versions of the computational model. For instance, the computational model we discuss here assumes that quantifiers have only one meaning threshold, which results in a monotonically increasing pattern for upward entailing quantifiers. However, it may be possible that the quantifier *Most* may have two meaning thresholds instead of one, if participants interpret it as *Most, but not all*. In this case, the observed response pattern would show a decrease in the proportion of ‘true’ responses for this quantifier for percentages close to 100. Since our computational model cannot capture this pattern, this mismatch would become apparent in the posterior predictive checks. The process of posterior predictive checks is illustrated in Figure 14.

To illustrate the importance of posterior predictive checks in our setup, consider a hypothetical scenario in which our computational model assumes no response noise among participants. Without the response noise parameter, the model assumes that participants respond ‘true’ only for presented percentages above their meaning threshold. The proportion of ‘true’ responses for presented percentages below the threshold is predicted to be zero at all times. However, people make errors when doing a truth-value judgment task (Ramotowska, 2022).

A comparison between real data, which includes response errors, and predictions from the model, which would be unable to account for these errors, can provide important insights into the true data-generating process. Figure 15 shows a hypothetical outcome of posterior predictive checks, based on a model without the response error parameter. The line represents the aggregated simulated predicted responses based on the posterior estimates, that is, the data the model predicts using the posterior samples. The triangles represent the observed data.

Below the estimated meaning threshold for this quantifier, the model exclusively predicts that individuals will never respond ‘true’, while the observed data shows that the proportion is non-zero. This illustrates that a model without the response error parameter cannot effectively capture the observed data below the threshold. These systematic deviations should prompt a refinement of the computational model. Overall, posterior predictive checks help us detect important information and challenge and update our theoretical assumptions (Schad et al., 2021).

Importantly, if posterior predictive checks lead to an adjustment of the model, any subsequent fitting to investigate a confirmatory research question must be based on new data. Fine-tuning a model with the final data leads to overfitting and is considered a questionable research practice (e.g., Pawel et al., 2024).

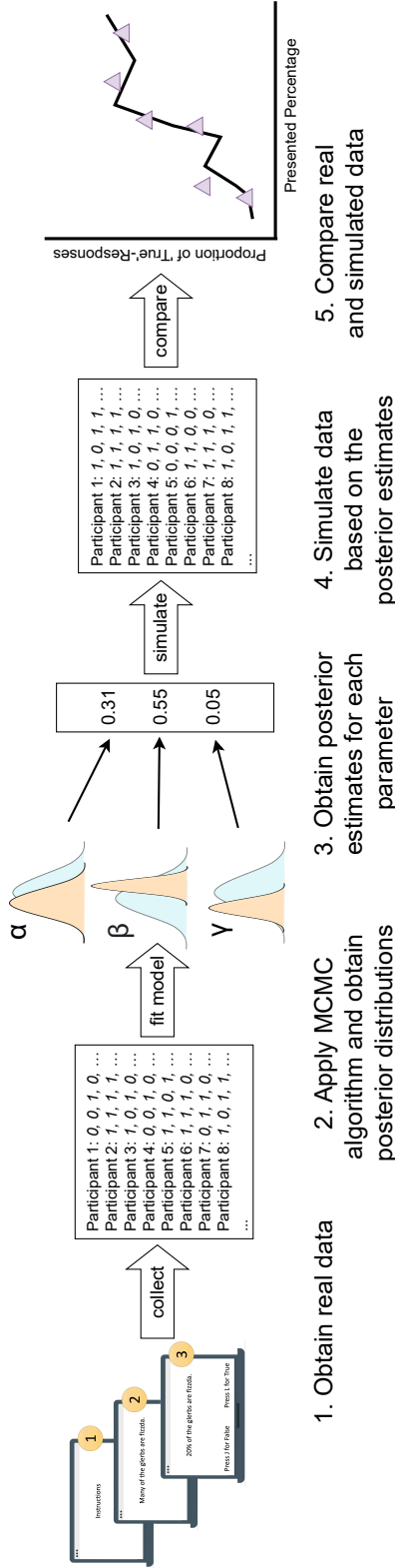


Figure 14: The process of posterior predictive checks for the purpose of illustration. For this, we need real observed data collected using our truth-value judgment task. We then fit the model to this data and obtain posterior estimates. Based on these posterior estimates, we simulate new data. Finally, the real observed and simulated data can be compared.

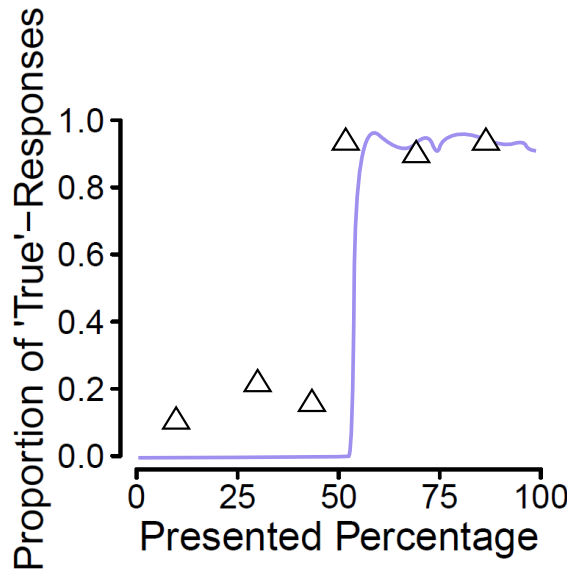


Figure 15: Hypothetical Posterior predictive check plot based on a model without the response error parameter. The line represents the aggregated simulated predicted responses based on the posterior estimates. The triangles represent the real observed data aggregated over 6 response categories. This illustrates that a model without response error parameter would not be able to capture the observed data below the threshold.

3.5.2 Application

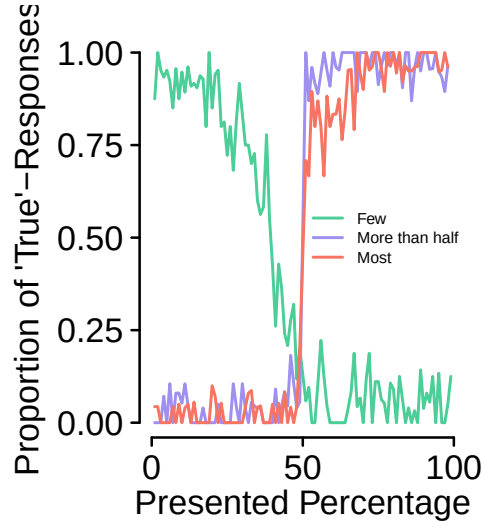
To obtain posterior predictions, we need an adjusted Stan model that predicts data based on the model fitted to the observed data. This updated Stan model is defined in *Chunk 24*. The adjusted model includes a generated quantities block that simulates datasets based on the fitted parameters. This enables us to compare the model-generated predictions with the observed data. The next chunk processes the data so that it can be passed on to Stan, followed by the actual sampling, post-processing and plotting (*Chunks 25-31*).

3.5.3 Interpretation

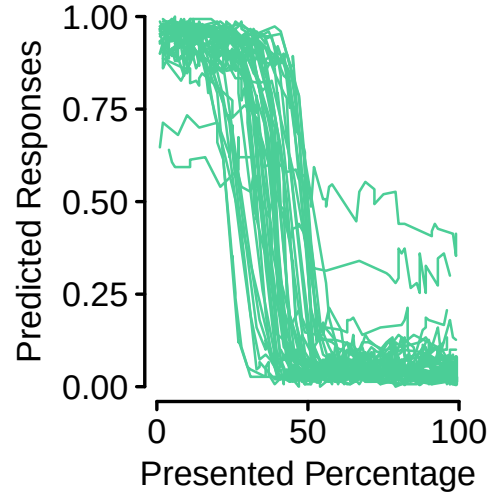
We discuss the posterior predictions for each quantifier and evaluate the overlap between the observed data and the model's predictions. Figure 16a presents the aggregated responses for all quantifiers in one plot. What is immediately apparent is the sharp transition from 'false' to 'true' answers for the quantifier *More than Half*, the increase in uncertainty around the threshold for the quantifier *Most*, and the greatest uncertainty for *Few*.

As with the prior predictions, we investigate the overlap both at the population level and the individual level. Figure 16 presents both the aggregated responses for each quantifier and the model predictions.

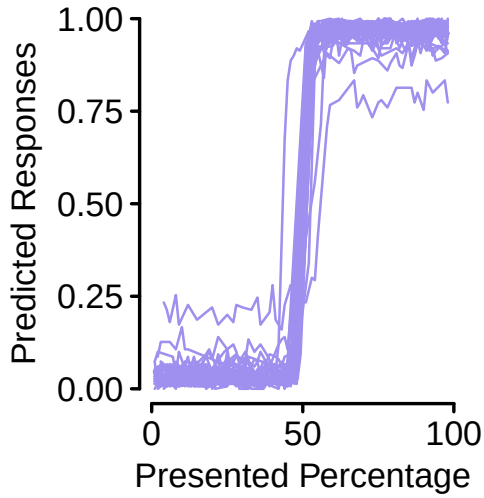
Few. The posterior predictive check for the quantifier *Few* is presented in Figure 16b at the population level, and in Figures 17a, 17d, and 17g for three randomly selected participants. At the population level, we see that the model is able to accurately redescribe the observed data. At the individual level, too, the model adequately captures the response behavior for participants 25 and 37. However, participant 1 exhibits an unexpected pattern: the observed data (displayed as triangles) suggest a response behavior opposite to what we would expect. Instead of a monotonically decreasing pattern, this participant displays a monotonically increasing pattern, meaning they are increasingly likely to consider high percentages as instances of the quantifier *Few*. Since our computational model cannot capture response patterns that violate the monotonicity assumption, it predicts data with very high response noise rates, nearly at chance level, resulting in a predicted response pattern that is almost a horizontal line.



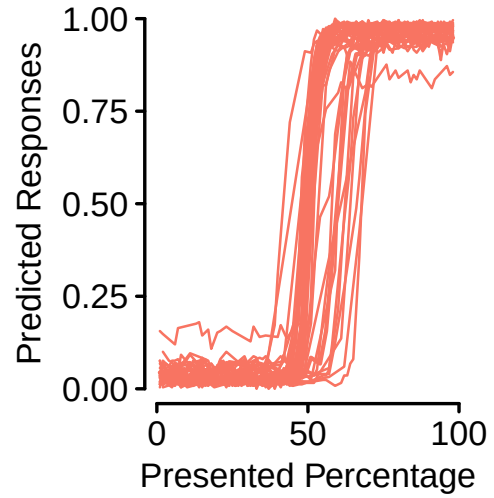
(a) Observed data for all quantifiers



(b) Quantifier: Few



(c) Quantifier: More Than Half



(d) Quantifier: Most

Figure 16: Panel a: The observed data for all quantifiers aggregated over the $n = 43$ participants. Panels b-d: Posterior predictives for all three quantifiers. These predicted data patterns are based on posterior draws from the model fit to real data, and printed for 43 participants.

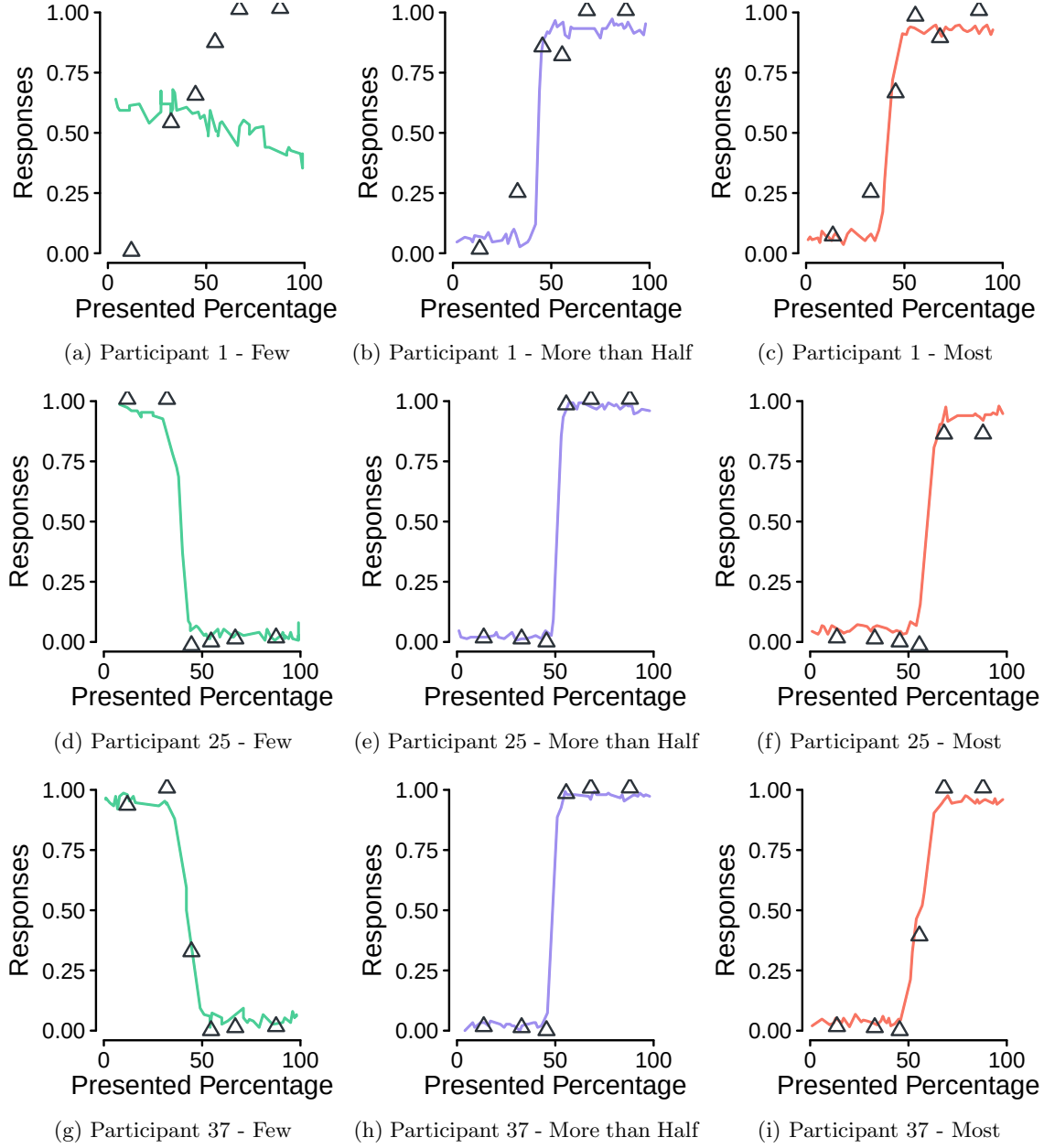


Figure 17: Posterior predictive checks for all three quantifiers for three randomly drawn participants. The line represents the aggregated simulated predicted responses based on the posterior estimates. The triangles represent the real observed data aggregated over 6 response categories. The percentage categories are 1 – 25%, 26 – 40%, 41 – 50%, 51 – 60%, 61 – 75% and 76 – 100%.

In this case, we do not attribute the mismatch between our computational model and the observed data to a failure of predictive adequacy. The response pattern is highly implausible; it is unlikely that an individual genuinely assigns a meaning to the quantifier *Few* that is completely opposite to what is logically sensible. Instead, it is more likely that this individual did not follow the task instructions properly. For instance, the participant may have ignored the quantified sentence and instead responded 'true' for higher percentages regardless of the quantifier.

Some modelers may feel inclined to exclude participants from the data if they clearly mismatch the posterior predictions. However, important exclusion criteria (e.g., excluding participants who violate the monotonicity assumption in this case) should be established prior to data collection and prior to fitting the model to the data. If modelers choose to exclude participants after the model has already been fitted to the data, subsequent interpretations and analyses should be interpreted with caution and flagged as exploratory, as this may also lead to overfitting.

More than Half. The posterior predictive check for the quantifier *More than Half* is presented in Figure 16c at the population level and in Figures 17b, 17e, and 17h for randomly selected participants. Overall, the observed response patterns align with the predictions from our computational model at both levels.

The proportions of 'true'-responses raise sharply for presented percentages above 50%, as expected for *More than Half*. We also observe some responses potentially influenced by response noise for participant 1, but no severe or systematic deviations in the subset of participants shown in this figure. Thus, for *More than Half*, we conclude that the model successfully captures the key aspects of the data-generating process.

Most. Finally, we turn to *Most*. The posterior predictive check for the quantifier *Most* is presented in Figure 16d at the population level and Figures 17c, 17f and 17i for randomly drawn participants. The response pattern for *Most* mimics that of *More than Half*, with the proportion of 'true' responses rising sharply above 50%, and participants overall exhibiting very low response noise. Here again, we do not observe any severe mismatch between the observed data and our model predictions, nor any extreme response patterns that challenge our theoretical assumptions. The pattern appears desirable for our third quantifier, so we can conclude that, here too, the model successfully redescibes the key aspects of the data.

Error Shooting Checklist

- Problem A: The computational model is not able to capture observed response patterns but instead shows systematic differences.
 - Solution A1: The model may not accurately reflect or capture the true data-generating process behind the observed data either at the group-level or the individual-level. In these cases, it is necessary to revisit the theoretical assumptions formalized in the computational model and assess whether they need to be adjusted. However, to avoid overfitting, it is essential to evaluate the refined model with new data.
- Problem B: The comparison between observed and simulated data shows no systematic but some differences for some individuals.
 - Solution B1: The model may fail to accurately describe the response pattern for individual participants if their behavior significantly differs from the rest of the participants. For hierarchical models, the estimated individual parameters can be shrunk towards the group mean due to hierarchical shrinkage, so that small deviations for outliers are not problematic, but rather desirable.
 - Solution B2: The model may fail to accurately describe the response pattern for individual participants if they violate model assumptions. This may happen if participants fail to follow the task instructions. If the suspicion of failure to adhere to the task instructions can be justified, the mismatch between observed and predicted data does not necessitate an update of the computational model.

Take Home Message

Posterior Predictive Checks. When we perform posterior predictive checks, we assess whether our computational model is capable of redescribing the observed data. This model validation check is conducted after empirical data has been collected and our model has been fitted to the data. Using the obtained posterior parameter estimates, we simulate new data and compare the observed and predicted data to identify systematic mismatches.

4 Concluding Remarks

In this tutorial, we applied various model validation techniques within a principled Bayesian workflow to a computational model that formalizes the semantic processes of quantifiers. We discussed the methods in detail and explained how to apply them. Additionally, we highlighted potential pitfalls and provided solutions for commonly encountered challenges. We also provided the R code to facilitate a step-by-step implementation while following the manuscript. While this tutorial is not a comprehensive treatment of the principled Bayesian workflow (Gelman et al., 2020), we focused on essential validation steps that, based on our experience, offer a clear understanding of the model’s validity, the computational algorithm, and the informativeness of the data.

Our results suggest that the presented Bayesian cognitive model performs well and can be used to study research questions related to the meaning threshold of quantifiers, the vagueness around the threshold and response noise. We employed the first model validation step with only 250 sets of samples from the prior distribution and the second and third model validation step with only 100 sets of samples from the posterior distribution. While a higher number of simulations might have provided even clearer results, this may not always be computationally feasible for complex models. However, even with this limited number, we were able to gain valuable insight into the performance of the model. With no clear signs of severe problematic patterns, we consider the validation process successful.

Our case study illustrated the key aspects of the data that a modeler may identify and how to constrain parameter priors to incorporate theoretical knowledge from the field. Naturally, when applying these validation steps to other semantic processes and research designs, these aspects will need to be adapted accordingly. We hope that this tutorial provides modelers in experimental semantics with an intuition and a roadmap on how to adapt the validation techniques to their specific cases.

To the best of our knowledge, although researchers in the field of experimental semantics already use Bayesian computational models to study semantic processes, validation studies are rarely conducted or performed. However, there is a real risk associated with the lack of validation. The prior distributions assigned to the model parameters may not reflect the beliefs of the modeler or the computational model may fail to effectively update the posterior distributions or even introduce systematic biases in the estimates; any conclusions drawn from such a model may be inaccurate or misleading.

At the same time, model validation steps offer valuable opportunities. Prior predictive checks allow us to clearly formulate theoretical assumptions and incorporate them into our model. Posterior predictive checks provide an opportunity to learn from observed data and update or challenge our theoretical assumptions. Thus, Bayesian computational models are not just tools for answering research questions; they are inherently linked to deep theoretical thinking.

Our goal was to make the application of a principled Bayesian workflow accessible and encourage modelers to adopt the techniques described here. We hope this tutorial enables researchers in experimental semantics to fully leverage their computational models and effectively integrate the field’s rich theoretical knowledge into their work.

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